Baltic Growth Puzzle: The First Latvian Soviet-Period Series and a Baltic Comparative Growth Analysis, 1919–2024

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Abstract

We provide the first annual estimates of Latvian GDP per capita for the Soviet period, 1937–95, built from physical output, transport and employment series drawn from Soviet yearbooks and archives and anchored to a 1990 purchasing-power-parity benchmark, and splice them to the existing interwar and modern series to obtain a continuous series for 1913 and 1919–2024 in international dollars. Independently constructed series agree within 3.5 per cent in every year of their overlap. The new series completes the Baltic record and permits the first national-accounts-based comparison of Latvia, Lithuania and Estonia under interwar capitalism, Soviet socialism and restored independence. Despite high social capability, none of the three converged on Western Europe before 1990: the growth of the 1930s and of 1950–68 was cancelled by the stagnation of 1968–89, and Latvia, the richest Baltic economy for most of the period from 1913 to 1988, is now the poorest. Catch-up came only after 1995, and has been conditional and uneven.

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INTRODUCTION

On the eve of the First World War a visitor to Riga or Tallinn found a 'culturally Western' province of the Russian Empire, where people worshipped in Lutheran churches, read in their own languages, traded with the West and enjoyed incomes closer to Europe's than to Russia's; the imperial government called the Baltic provinces 'our little West' (Corum 2018, 129; Kodres 2021, 427). In 1913 income per head in Latvia and Estonia stood at the level of Finland or Italy, and even agrarian Lithuania was richer than Greece or Portugal (Norkus 2023c, 129–30). When Red Army soldiers entered the Baltic republics under the mutual-assistance pacts of 1939 they wrote home in astonishment: 'the store is full of all sorts of fabrics, whatever the soul desires, and there is not a single person, not to mention a line' (Mel'tiukhov 2014, 279). By 1990, after forty-five years inside the Soviet Union, income per head in the three republics was, on the series presented here, about half that of Finland and three-fifths that of the Western periphery; by 2024, after three decades of reform, European Union membership and 'Baltic tiger' growth, they remained in Europe's periphery, and the richest of them in 1913, Latvia, had become the poorest, as World Bank figures for 2022 already showed (Köll 2025, 93). Three small, literate, egalitarian economies, which possessed from the outset much of what the catch-up literature regards as the social capability needed to grow (Abramovitz 1986, 388), have failed over 110 years and under three economic regimes to leave the periphery into which the nineteenth century placed them. That is the Baltic growth puzzle.

The puzzle has not been posed in these terms because the data to pose it did not exist. The Maddison Project Database, the standard source for long-run comparisons, reports GDP per capita for Lithuania, Latvia and Estonia only for 1973 and from 1980, and its pre-1990 figures descend from Soviet net material product, which Maddison himself regarded as unusable (Maddison 1998, 307–8; Bolt and van Zanden 2025). The Baltic states have accordingly been absent from the economic histories of twentieth-century Europe, and discussion of their living standards has been left to memory and politics: a romanticised interwar 'golden age' on one side, and on the other the image of the Soviet republics as impoverished and exploited colonies (Lehti 2005, 92; Köll 2025, 98). The gap has begun to close. Direct benchmark estimates now exist for Lithuania in 1937, Latvia in 1935 and Estonia in 1938, annual interwar series have been

built on them, and the first Soviet-period series, for Lithuania and Estonia, have shown that the Maddison Project's late-Soviet levels are too high by between a sixth and a third (Klimantas and Zirgulis 2020; Norkus et al. 2024; Klimantas et al. 2024; Klimantas 2026; Klimantas and Klesment 2026). Latvia is the missing piece. Its interwar record is the most disputed of the three, and for the Soviet period there is no estimate of real output at all: the Western Sovietologists who studied it worked from official aggregates (King 1965; Viksnins 1981), and Latvian historians, who have reconstructed the institutional history of the Soviet economy in detail, have had to rely on official indices (Krūmiņš 2017g, 277–8) whose growth, on the present estimates, is several times the real one.

This paper makes three contributions. First, it provides the first annual series of Latvian GDP for the Soviet period, 1937–95, built from 129 physical product series, agricultural output, transport volumes and employment records drawn from Soviet yearbooks, the German occupation statistics of 1942–4 and the Latvian and Russian archives, anchored to Bolotin's purchasing-power-parity benchmark for 1990 with the methods of the CIA's accounts for the Soviet Union and of the companion Lithuanian and Estonian series. Second, it splices the new series to the published interwar series, extended here to 1919 and 1940, and to adjusted post-1991 data, to give a continuous series for 1913 and 1919–2024 in 1990 and 2011 international dollars, with a consistent population series, ready for the Maddison Project Database. Because the interwar and Soviet-period series are built from different sources and anchored to benchmarks fifty-five years apart, their agreement where they overlap, within 3.5 per cent in each of the years 1937–40 and within 3 per cent at the 1990–1 splice, is the strongest available check on the whole reconstruction. Third, and principally, it uses the completed Baltic record to provide the first comparison of the three Baltic economies, with each other and with the rest of Europe, that rests on national accounts rather than proxies, across interwar capitalism, Soviet socialism and restored independence. The Latvian series is analysed briefly, since the companion papers have examined the Lithuanian and Estonian cases in depth; the weight of the paper falls on the comparison.

The new series alters the received picture in four respects. First, the Baltic states did not converge on Western Europe under interwar capitalism. Latvia emerged from the First World

War as perhaps the most devastated economy in Europe (Krūmiņš 2024, 44; Norkus 2014, 323), and on the present series both it and Lithuania had lost, like Russia, three-fifths of their output, with income per head more than a third below 1913; the 1920s were spent regaining it, and the rapid growth of 1929–38, 3.1 per cent a year in Latvia and almost as much in its neighbours, faster than anywhere in Europe outside Finland and the Soviet Union, only restored the region to 49 per cent of the Western core, where it had stood in 1913. Second, the Soviet period divides in two. Between 1950 and 1968 income per head grew by 4.0 per cent a year in Latvia and faster in Lithuania, and the region reached 62 per cent of the Western core, closer than at any time before 2019; between 1968 and 1989 Latvia grew by 0.7 per cent a year, slower than any economy of the communist bloc for which a series exists, and the gains were lost. Within the Union the Baltic republics had the highest living standards at the end of the Soviet period (Hiden and Salmon 1994, 127), Latvia and Estonia the highest GNP per head in 1988 (van Selm 1997, 35), and on the new series Latvia and Estonia stood well above the Soviet average throughout, Latvia at 159 per cent of the Soviet average in 1968 and 136 in 1989; the lead was inherited rather than created by Soviet investment, which as a share of the republic's GDP stayed below the Union average, and it narrowed in the final decades. Third, the transition cut Latvian income per head by 45 per cent in four years, the deepest of the three Baltic contractions, and the convergence that followed, at 8.5 per cent a year to 2007, is the only sustained convergence the region has known; since 2007 it has been uneven, and Lithuania, the poorest of the three for most of the century, overtook Latvia durably in 2010 and Estonia in 2018. Fourth, the belief that the Baltic states stood level with Finland before the war, which underlies most estimates of the cost of occupation (Kalniņš and Piņķe 2017, 60–2; Kukk 2005, 167–8; Viksnins 1986, 252), is not borne out: Latvia had fallen from 112 per cent of the Finnish level in 1913 to 86 per cent by 1938, before Soviet rule began.

The paper also speaks to a wider literature. The economic history of the Soviet Union has been rewritten from archival sources, but that work has concentrated on Russia, and the regional development of the Union, which Kukić has called a major gap in its spatial economic analysis, remains almost without national accounts (Zhuravskaya et al. 2024; Kukić 2024, 670). The Baltic states, with borders stable since 1945 and a capitalist past against which the socialist

record can be measured, offer the first regional evidence, and the procedure used here can be applied to Ukraine, Belarus and the other republics. They also add three cases to the debate on peripheral convergence in Europe, in which Eastern Europe's failure to catch up under socialism (Vonyó and Markevich 2021, 282) and its long-run backwardness, traced to a semi-feudal past (Berend 2020, 42–3), have been much debated, while for the interwar years Morys (2021, 163) finds that the region began to converge. The Baltic states had shed that semi-feudal past by 1925; their failure to converge before 1990 therefore points to the regimes and the external environment rather than to the societies.

The next section reviews the data gap and the earlier estimates. The following section describes the sources and methods of the Latvian Soviet-period series, its splicing to the interwar and modern series, and the limits of the result. A brief section then reads the Latvian series. The core of the paper compares the three Baltic economies under the three regimes, with growth rates, relative levels, structure, convergence tests and a discussion of the sources of catch-up and decline. A final section concludes.

THE BALTIC DATA GAP AND PREVIOUS ESTIMATES

The absence of the Baltic states from the quantitative economic history of Europe is a consequence of their history rather than of neglect. Latvia and Estonia have a remarkably brief record of state continuity, so that little impetus existed before 1918 to maintain the consistent statistical records that state-building elsewhere produced; the Russian Empire into which all three future republics had been absorbed in the eighteenth century was itself a country of weak statistical coverage (Gregg 2024, 219–20). Regular statistical publication began with independence, and Latvia was among the first ten states in the world whose statistical office published estimates of national income, for the mid-1930s (Norkus et al. 2024); its interwar statistical service was among the best in the Baltic states, and it computed and published national income for the 1930s (Norkus 2014, 112). But the period from 1940 to 1945 saw the Baltic states occupied three times, with changes of borders, economic regime and statistical practice and, in Lithuania at least, the physical destruction of part of the archival record (Dagys 1959, 140), and after incorporation the economic history of the independent decades was systematically neglected (Morys and Ivanov 2021, 244). For the Soviet period the difficulty is of a different kind. The

republics' 'national income' statistics, compiled from the 1950s, measured net material product, which excludes most services and is net of depreciation; the ratio of net material product to GDP varies across countries and over time, so that conversion is unsafe (Češka 1989, 581–93; Campbell 1985); and the series were compiled at 'comparable prices' that understated inflation and overstated growth (Mertelsmann 2003, 164–7; Misiunas and Taagepera 1993, 289), a practice that Adlys, deputy head of the Lithuanian statistical committee, acknowledged in a published interview of 1988: five-year plans 'were not fulfilled but we showed that they were' (Adlys 1988). The official index of Latvian industrial production at constant prices rose nineteen-fold between 1940 and 1980 (Misiunas and Taagepera 1993, 361), more than twice the increase recorded by the physical-product index built in the next section; its 1940 base, moreover, values the last pre-Soviet year at the official lat–rouble rate, which a Soviet memorandum of July 1940 reckoned undervalued the lat six- to eight-fold (Terleckas 2022, 290), as the official rate undervalued Estonian production almost five-fold (Mertelsmann 2003, 167), and in physical units the 1937 level had not been regained by 1947 (Misiunas and Taagepera 1993, 109).

Neither the Baltic statistical offices nor Eurostat, the IMF or the World Bank publish GDP for the Baltic states before 1990, and the World Bank's own reconstructions of 1980–91, made by its missions of 1992–3 (World Bank 1993, 4; The World Bank 1995, 32–3), were not continued. The Maddison Project Database reports Latvian GDP per capita for 1973 and annually from 1980. Its 1990 level descends from Maddison's application of Bolotin's republic-to-Union ratios, taken for 1991 rather than 1990, to a Soviet total derived from a bilateral Austria–USSR comparison; its earlier years are extrapolated backwards with net material product growth adjusted by the Union-wide ratio of GDP to net material product (Maddison 2006, 181–2), although Maddison himself held the official Soviet growth statistics to be unacceptable, since they 'overstated growth, and understated inflation' (Maddison 1998, 307–8). The resulting series shows Latvian income per head rising by 5 per cent in 1989, when every physical indicator was already falling, a rise inherited from the official series and reproduced in the EBRD's compilations (Kyn 1997, 269), which also show growth in 1990 (Kukk 2014, 13), and stands 24 per cent above Bolotin's independent benchmark for 1990 (Table 5). The only republic-level estimates in the Maddison tradition are the indirect approximations of Norkus and Markevičiūtė (2021) for 1913, 1922,

1929 and 1938, obtained from real wages and the structure of employment, and the figures of Rosés and Wolf (2010, 187–90) for the same interwar years, which rest on an undocumented procedure and make interwar Latvia the fastest-growing economy in Europe outside the Soviet Union and, by 1938, richer than Finland; the authors themselves concede that the estimates 'might be on the high side'.[1]

For Latvia the direct estimates are recent and confined to the interwar years. Norkus et al. (2024) produced an output-side benchmark for 1935 within the 2008 System of National Accounts, with gross value-added for twenty industries at basic and market prices and a conversion into international dollars by price comparisons with two neighbouring countries, Sweden and Lithuania (Klimantas et al. 2024); Klimantas et al. (2024) extrapolated it to annual series for 1920–39 from sectoral volume indices, resolving the long dispute between Aizsilnieks, for whom national income per head did not grow under the Ulmanis dictatorship of 1934–40 (a reading Karnups has lately revived in part, finding that national income grew only slowly and regained its 1929 level only in 1938 (Karnups 2017a, 145, 148–9)), and the Latvian historians who regarded those years as prosperous (Krūmiņš 2017b, 26): growth continued after 1934, but Latvia was not Europe's fastest-growing economy in 1929–38,[2] and its re-industrialisation was mainly import-substituting (Aizsilnieks 1968, 790, 834; Valge 1994, 188). The series is the anchor of the present paper. For the Soviet period, whose later decades remain largely a blank spot in Baltic historiography (Köll 2025, 98), there is no comparable Latvian estimate. The Western Sovietologists who studied the Latvian economy had no access to planning data and worked from published Soviet sources (Grybkauskas 2022, 485), which they could at best qualify (e.g. Misiunas and Taagepera 1993, 289): King's study of economic policy in occupied Latvia and Viksnins's evaluations of Latvian growth under Brezhnev and Gorbachev document the structure of investment, labour and consumption but offer at most rough dollar comparisons (Viksnins 1986, 238–9) and no measure of real output comparable across time or countries (King 1965; Viksnins 1981), and Koropeckyj's attempt to place the national income of Latvia

1. Rosés and Wolf also attribute the Baltic recovery to 'an early exit from the gold standard with the UK in late 1931' (Rosés and Wolf 2010, 189); Latvia in fact imposed exchange control in October 1931 and left gold only on 28 September 1936 (Bernanke 2000, 74; Richter 2026, 146).

2. Bukowski et al. (2026, 3, 19–20), using the same Latvian series, call Latvia 'the region's and Europe's leader of economic growth' for 1924–38, at 3.4 per cent a year; their comparison set excludes Finland and the Soviet Union.

and Estonia in 1970 at British and Australian levels illustrates the hazards of such comparisons (Koropeckyj 1976, 68, 72), as does Fedotov's claim of 1988 that Latvian national income per head exceeded 70 per cent of the American level (Kukk 2005, 166). Latvian historiography since 1991 has produced a rich institutional account of the Soviet economy, culminating in the collective economic history edited by one of the present authors and in the reassessment of Latvia's fiscal relations with Moscow, which overturned the Soviet claim that the republic was a net recipient of Union investment (Krūmiņš 2017g; Krūmiņš 2018), a conclusion consistent with the Baltic share of Union investment in 1946–60, which was below the region's share of output (Vaskela 2014, 233); but it, too, has had to rely on Soviet output indices whose growth far exceeds the real one. The Lithuanian and Estonian series now available close two-thirds of the gap (Klimantas 2026; Klimantas and Klesment 2026). The Latvian series presented in the next section closes the rest, and allows the three Baltic economies, which are compared with each other more often than with anyone else, to be compared on a common footing for the first time.

ESTIMATING LATVIAN GDP, 1937–1995

The output approach and the reliability of Soviet statistics

For the Soviet period only the production side of national income can be reconstructed. The monetary aggregates that an expenditure or income estimate would require are the figures that cannot be trusted: the Material Product System measured net material product at administered prices, excluded most services, double-counted intermediate transactions in its gross-output measures and, above all, understated inflation, so that official constant-price growth was too high; in Maddison's verdict, the official series 'overstated growth, and understated inflation' (Maddison 1998, 307). The Western response, begun by Bergson in the 1950s and institutionalised by the CIA from the 1960s, was to rebuild Soviet GNP from the bottom up, re-pricing output at factor cost, imputing the missing services and replacing the official value indices with indices built from physical output series weighted by value-added (Bergson 1950, 1961; Joint Economic Committee 1982). The CIA estimates were attacked from both sides, by critics who held that they overstated Soviet growth by ignoring deteriorating quality, over-reporting and hidden inflation, and faulted even the physical series as 'open' statistics compiled for external consumption, and

by a minority who contended that military output, housing and services were under-measured (Navrozov 1988; Becker 1994; Schroeder 1995); Khanin's alternative statistics, as Harrison (1993, 144, 159) showed, coincided with the CIA's for 1950–87. The assessments of Becker and Schroeder and the review by Maddison conclude that the CIA series contain errors in both directions, that most of the criticism 'was not well founded', and that the physical-output method remains the best basis for measuring Soviet growth; the CIA's estimates, in their final version at 1982 prices (Joint Economic Committee 1990), are the basis of Maddison's Soviet series, and Vonyó and Markevich describe the Western revisions as still widely acknowledged as the best available (Maddison 1998, 309; Vonyó and Markevich 2021, 280). Ponomarenko's archival reconstruction of Russian national accounts reaches the same judgement from inside the Soviet statistical system: the physical output data 'are apparently of high quality and can be used', whereas the official value indices were inflated by the pricing of new products (Ponomarenko 2002, 16–18, 73–4). The literature on the Soviet economy has grown rapidly since the opening of the archives (Zhuravskaya et al. 2024), but its quantitative core for the post-war decades still rests on physical output series of this kind, and the non-Russian republics remain almost entirely without modern national accounts, studies of Soviet growth being conducted at the level of the Union or of Russia (Kukić 2024, 646, 670–1; Voskoboynikov 2021).

The series presented here follows this tradition and the procedure used for Lithuania and Estonia (Klimantas 2026; Klimantas and Klesment 2026). A benchmark estimate of GDP is extrapolated with constant-price indices of sectoral value-added, each sector weighted by its share of GDP in the benchmark year:

$$GDP_t = GDP_{\text{bench}} \times \sum_{i=1}^{n} w_i \, I_{it},$$

where w_i is sector i's share of benchmark GDP and I_{it} its volume index in year t. The indices track gross output, inputs or employment and are combined with value-added weights, what Thomas and Feinstein call the indicator approach, as distinct from single deflation, although like it the procedure holds the ratio of value-added to output constant; it is the procedure that Fabricant, as they quote him, called the 'next best' where double deflation is impossible because intermediate inputs cannot be observed (Thomas and Feinstein 2003, 427–8, 444), and

the CIA itself double-deflated only agriculture (Joint Economic Committee 1982, 45). It matters, therefore, which Soviet data are used and which avoided. The distortions documented in the literature were concentrated in value aggregates, in the national income and gross output indices at 'comparable prices', in the reporting of hidden price increases as real growth and in the double counting of the material product system (Maddison 1998, 307–8; Vonyó 2017, 6; Mertelsmann 2003, 165); Khanin's own classification, as Harrison summarises it, is that 'the ease of distortion remained greater for value series than for physical volumes', and that distortion was concentrated in heterogeneous products such as machinery and construction, while series for grain, milk, cement, electricity or freight are comparatively reliable (Harrison 1993, 144, 152). The two diverge widely: the official Soviet index of Latvian industrial output at constant prices rose nineteen-fold between 1940 and 1980 (Misiunas and Taagepera 1993, 361), more than twice the increase recorded by the physical-product index compiled here (Table 7). The official index also values its 1940 base at an arbitrary lat–rouble parity that undervalued the last pre-Soviet year several-fold (Misiunas and Taagepera suspected in addition that only the five months of Soviet rule were counted; Mertelsmann finds for Estonia that the whole year was); its claim that industry had surpassed 1940 by 1947 is contradicted by the physical series (Misiunas and Taagepera 1993, 109; Mertelsmann 2003, 167; Krūmiņš 2017f, 234). The indices below rest almost entirely on physical quantities, on employment and on inputs; the one place where value data at comparable prices enter, trade turnover, which also scales the finance index, is flagged and its bias bounded in the sensitivity analysis.

The 1990 benchmark

The Soviet-period series is anchored to Bolotin's estimate of the GDP of the Latvian SSR in 1990, which expresses the accounts of all fifteen republics in international dollars (Bolotin 1992, 181–92). The year 1990 is one of the first for which republican accounts on an SNA-type definition were compiled (Guļāns 2015, 43–4) and the first with purchasing-power parities; Bolotin converted the republican GDP of that year into dollars with parities estimated, as he put it, 'in practically full conformity' with the International Comparison Project. The parities, by sector of origin, were 0.40 dollars per rouble in agriculture, 1.80 in industry, 1.10 in construction, 2.80 in transport, 1.45 in trade and 3.30 in other services, or 1.82 for GDP as a whole (Bolotin 1992,

181, 184). For Latvia the result is 7,965 Int$$_{1990}$ per head, between Lithuania's 7,475 and Estonia's 8,425. Table 1 shows the sectoral composition in roubles, with services other than trade and transport allocated by the 1987 Soviet input–output table (Steinberg 1992), beside the 1935 benchmark of Norkus et al. (2024) on which the interwar series rests. The rouble shares are domestic-price weights, like the lats shares of 1935, and are the weights with which the volume indices are combined; the dollar conversion applies to the total alone. They are at the market prices of 1990, so that product taxes and subsidies are carried within the sectors that bore them; no re-pricing to adjusted factor cost was attempted, since no republican data on the incidence of net product taxes by branch exist, and the only socialist economies for which the World Bank reports the ratio of product taxes less subsidies to value-added, Bulgaria and Ukraine in 1980–90, show about 0.5 per cent (The World Bank 2024).[3]

Author check (Table 1). The 1990 industry value, 3,190.8 million roubles, does not follow from Bolotin: his 5.2 billion dollars at the industry parity of 1.80 give 2,888.9 million roubles, while every other sector reproduces exactly (Bolotin 1992, 184). The dissertation's benchmark table carries the same uplift (+301.9 for Latvia, +307.2 for Lithuania, +259.0 for Estonia). State the adjustment here and in the source note of Table 1, or revert to Bolotin's figure.

Three features of the benchmark need stating, because the reconstruction stands or falls with it. First, its level is conservative. Bolotin's figure for the Soviet Union as a whole, 6,275 Int$$_{1990}$ per head, lies below Maddison's own estimate for 1987, and the bilateral Austria–USSR comparison on which it draws placed the Union at 32 per cent of the United States where the CIA had placed it at 42 per cent (Maddison 1998, 318–19). Maddison nevertheless built the Maddison Project's 1990 levels for the republics from Bolotin's republic-to-Union ratios of 1991 rather than 1990, ratios inflated by the first year of transition, and then carried them backwards with net material product growth (Maddison 2006, 181–2); the consequences for Latvia are shown in Table 5 below. Second, the parities are Union-wide, so that the conversion assumes that relative prices in Latvia were those of the Soviet Union as a whole. For the material sectors this is a reasonable approximation in a planned economy with administered prices; for services, where the parity of 3.30 dollars per rouble reflects the very low rouble wages of the non-market sector,

3. The turnover tax, the principal product tax, was levied on final consumption goods and accrued to the republic where the finished good was made, as with cotton fabric woven in the Baltics from cheaply procured Uzbek cotton (Yasin 1993, 20).

Table 1

Benchmark GDP of Latvia by sector of origin, 1990 and 1935, current prices

	1990		1935	
Sector	m. roubles	%	m. lats	%
Agriculture, forestry and fishing	2,000.0	17.6	383.4	36.1
Industry	3,190.8	28.0	216.5	20.4
Construction	818.2	7.2	23.6	2.2
Transport and communications	964.3	8.5	53.2	5.0
Trade, accommodation and catering	1,448.3	12.7		
Finance and insurance	234.1	2.1		
Housing	689.2	6.1		
Commercial services, 1935 total			*233.6*	*22.0*
Education and culture	580.2	5.1		
Health	357.1	3.1		
Science	203.4	1.8		
Public administration	114.8	1.0		
Defence	626.9	5.5		
Public services, 1935 total			*103.5*	*9.7*
Other services	163.9	1.4	48.0	4.5
Gross value-added	11,391.2	100.0	1,061.8	100.0
Taxes less subsidies on products	—		77.8	
GDP at market prices	11,391.2		1,139.6	
GDP per capita, national currency	4,277.3		583.4	
GDP per capita, Int$$_{1990}$	*7,965*		*2,776*	

Notes: the 1990 benchmark is at market prices: taxes less subsidies on products are carried within the sectors that bore them and are not reported separately (shown as '—'). The 1935 benchmark reports commercial and public services as two aggregates, shown in italics. Population: mid-year. *Sources:* 1990: Bolotin (1992, 184–6), converted back to roubles with Bolotin's sectoral purchasing-power parities, services other than trade and transport split with the 1987 Soviet input–output table (Steinberg 1992); 1935: Norkus et al. (2024, 263–6, 287, 312); per capita figures in Int$$_{1990}$: see online appendix.

it is less certain, and the resulting share of services in Latvian GDP at rouble prices, 47 per cent, is above the 34 per cent implied by the World Bank's 1990 sectoral data (The World Bank 1994); the sensitivity of the series to this allocation is examined below. Third, there is no alternative.[4] Ponomarenko's accounts cover Russia alone and make no dollar conversion; the World Bank's GDP levels for 1980–91 (World Bank 1993; The World Bank 1995, 32–3) were not continued; the UNECE's 'real GDP' indices for 1980–90 were taken from net material product (United Nations Economic Commission for Europe 1995, 249); the CIA's 1982 benchmark cannot be decomposed by republic, and the 1987 input–output table understates services too severely to

4. Earlier claims ranged widely: PlanEcon's CIA-based estimate for Estonia in 1989 was 6,240 dollars against 5,000 for the Union (Kukk 2005, 166).

serve as a benchmark (Steinberg 1992).

The choice of 1990 means that the series is extrapolated backwards for more than fifty years from end-of-period weights. The practical reason is that 1990 is the only year for which republican accounts on an SNA definition and purchasing-power parities exist together. Two further considerations favour it: the reporting of 1989–90, at the height of glasnost, was more transparent than that of any earlier Soviet year, the manipulation of earlier returns having been publicly acknowledged in 1988 by the deputy head of the Lithuanian statistical committee (Adlys 1988); and 1990 is a year in which the branches built up after the war, chemicals and pharmaceuticals at Olaine, electrical engineering or microbus production, can be observed and weighted at all (Krūmiņš 2017d, 278; 2017e, 303, 305). The cost is that half a century is valued at end-period relative prices, which, as the sensitivity analysis explains, tends to understate growth; the short answer is that the consequence can be tested directly, because the Soviet-period series meets an independent series with 1935 weights in 1937–40.

Volume indices and their sources

Table 2 summarises the indicators and sources for each sector; the online appendix documents every series year by year. The Soviet-period indices are built from the statistical yearbooks of the Latvian SSR, which from 1957 give annual physical output from the mid-1950s and, unusually among the Baltic republics, cover the early post-war years, though incompletely; from the statistical bulletins published by the German civil administration for 1942–4; from the USSR compendia on domestic trade and on the national economy; from the World Bank's studies of 1993–6; and from archival material, notably the returns of the Latvian statistical administration on sown areas for 1945–55 and the transport returns of 1945–50 held in the Russian State Archive of the Economy. Three general points precede the detail. First, archival material is used to fill the early post-war years that the yearbooks cover incompletely, not merely to decorate the series; but, unlike for Lithuania and Estonia, no archival compilation of late-1980s output values at the product level could be found for Latvia, so that the 129 product series are consolidated into 22 minor branches by averaging and the minor branches weighted with the 1987 input–output study and the World Bank's 1993 valuation (Steinberg 1992, 82–5; World Bank 1993, 289). Second, the series cover the civilian economy as reported to the republican statistical office.

Military production, organised through all-Union enterprises and never reported separately, is not captured, and Latvia hosted more of it than its neighbours: the VEF electrotechnical works, the Riga radio and instrument plants and the shipyards all carried 'special' production for the Soviet armed forces (Upmalis 2015, 59–61; Blaubergs 2015, 74–5; World Bank 1993, 103–4); the size of the Soviet garrison is likewise unknown between 1941 and 1989 and is treated as a source of bias below. Third, the year 1944 proved beyond reconstruction from Latvian sources, so that sectoral movements for 1943–5 are taken from the ratios observed in Lithuania, where documentation is better.

Table 2
Indicators and sources of the Latvian Soviet-period series, 1937–95

Sector	Indicator and main sources	Appendix
Crops	Physical output of 9–10 crop categories, 1937–95; 1990 value-added weights from the World Bank's 1993 study, grains split with the 1995 Eurostat agricultural accounts; sown areas for 1945–55 from unpublished returns	A.1
Livestock	Physical output of 7–16 animal products; 1941–9 yields obtained from animal numbers and the yields per animal of 1940, 1945 and 1950	A.1
Forestry and fishing	Timber cut (cubic metres) and fish catch (tonnes); 1941, 1971 and 1974–6 catches interpolated	A.1
Industry	129 physical product series in 11 branches (Table 3); product weights from the 1987 input–output study and the 1993 World Bank study, products consolidated into 22 minor branches by index averages because no archival late-1980s output valuation survives for Latvia; branch weights from the 1990 benchmark	A.2
Construction	Weighted index of material inputs into construction, 1987 input–output weights, following the CIA procedure	A.3
Transport and communications	Freight tonne-kilometres and passenger-kilometres, all modes; postal, telegraph and telephone traffic and radio and television sets; 1945–50 from the Russian State Archive of the Economy; 1941–4 assumed to move with the rest of the economy; 1990s from sectoral employment	A.4
Trade, accommodation and catering	Soviet retail and wholesale turnover indices at comparable prices (the only value series used), chained across base-year changes; early post-war years from the USSR domestic-trade compendia	A.5
Finance, insurance and housing	Finance and insurance: employment by branch, adjusted for productivity with the ratio of trade turnover to trade employment, as the CIA did; housing: housing stock derived from construction and population	A.5
Public administration, health, education, science	Employment by branch from Soviet yearbooks, the Ostland statistical bulletins and World Bank studies; 1941–4, 1946–9 and 1951–4 gaps interpolated	A.6
Defence	Soviet garrison of 30,000 in 1940 and 80,000 in 1990 plus the republican forces and police of 1940; 1941–89 interpolated between the 1940 and 1990 totals in step with public-administration employment; 1991–5 from the withdrawal schedule and the new Latvian forces	A.6
Other services	Employment	A.6
Population	Year-end estimates and censuses of 1935, 1959, 1970, 1979 and 1989; wartime and deportation losses from the demographic literature; Soviet garrison added in 1940–1; present-day borders from 1945	A.7

Notes: the appendix column refers to the sections of the online appendix in which every series is documented year by year with its sources: the Soviet Latvian statistical yearbooks (*Latvijas PSR tautas saimniecība*, 1957–90), the *Statistische Berichte für das Ostland* (1942–4) and *Ostland in Zahlen* (1942), the USSR domestic-trade compendia (*Sovetskaya torgovlya*, 1964 and 1989), the World Bank's *Latvia: the transition to a market economy* (1993) and *Statistical handbook: states of the former USSR* (1994–6), the Russian State Archive of the Economy (RGAE) and the Latvian State Archives. *Sources:* see online appendix.

Agriculture, forestry and fishing. The index combines physical output of nine to ten crop categories and seven to sixteen animal products with timber removals and the fish catch, each sub-index weighted by its 1990 value-added, the crop and livestock weights coming from the World Bank's 1993 study and the split within grains from the Eurostat agricultural accounts

Table 3
Industrial product series compiled for the Soviet-period indices, 1937–95

Branch	Latvia	Lithuania	Estonia
Electricity	1	1	1
Petroleum	0	1	3
Other fuels	3	1	2
Metals	4	7	4
Chemicals	8	10	11
Machine-building and metal-working	30	36	21
Wood and paper	9	14	10
Construction materials	22	15	8
Light industry	18	18	9
Food industry	25	36	25
Other	9	4	7
Total	129	143	101

Sources: this paper; Klimantas (2026); Klimantas and Klesment (2026).

(World Bank 1993, 286). Early post-war output comes from unpublished returns on sown areas; yields for 1941–9 are obtained from animal numbers and the yields per animal of 1940, 1945 and 1950; disaggregated grain data for 1947–52 are not available, so that the grain categories move with the total.

Industry. Physical output series could be assembled for 129 products in ten of the eleven branches of the benchmark, electricity, fuels, metals, chemicals, machine-building and metal-working, wood and paper, building materials, light industry, food and a residual; the eleventh, petroleum refining, did not exist in Latvia (Table 3). The fuel index for 1980–90 is taken from the World Bank's study and extended with peat and natural gas; where a product series begins or ends with zero, production is assumed absent in the adjacent years. The branches are combined with the branch structure of the 1990 benchmark. For the interwar years the published manufacturing volume index for 1930–9 is used and the earlier years rest on employment adjusted for real wages, as in the Lithuanian and Estonian series (*Ostland in Zahlen* 1942, 39, 45).

Construction. The Soviet-period index is a weighted index of the material inputs into construction, with weights from the 1987 input–output table, which is the CIA's procedure (Steinberg 1992, 66–89).

Transport and communications. Freight tonne-kilometres and passenger-kilometres across all modes are combined with a communications index built from postal and telegraph traffic, long-distance calls and radio and television sets in use, with weights from the 1987 input–output

study. The years 1945–50 come from the Russian State Archive of the Economy; for 1941–4 no transport data exist and the sector is assumed to move with the rest of the economy; after 1990 the index follows sectoral employment.

Trade, finance and housing. Under planning industrial products circulated by allocation rather than through markets, so that the marketed-share approach of the interwar series cannot be used. The Soviet-period trade index instead chains the retail and wholesale turnover indices at comparable prices compiled by the Soviet statistical offices, the early post-war years coming from the USSR domestic-trade compendia; these are the only value series in the reconstruction. Finance and insurance follow employment, as in the CIA's estimates, which used employment as a proxy for output in such sectors (Maddison 1998, 319; Joint Economic Committee 1982, 49); unlike the CIA, which found no evidence on which to base a productivity allowance, the present series adjusts employment for productivity with the ratio of trade turnover to trade employment. Housing follows the housing stock, as in the CIA's index (Joint Economic Committee 1982, 46), here derived from construction and population.

Public and other services. Public administration, health, education and science follow employment by branch, as in the CIA's accounts, with gaps in 1941–4, 1946–9 and 1951–4 interpolated. Defence is the least secure component. The Soviet garrison admitted under the mutual-assistance pact of October 1939 numbered about 30,000 (Misiunas and Taagepera 1993, 15–16), and some 80,000 Soviet troops were stationed in Latvia in 1990, far more in proportion to population than in Lithuania or Estonia; between those dates the garrison is interpolated between the 1940 and 1990 totals in step with public-administration employment, and after 1990 the index follows the withdrawal schedule and the formation of the Latvian forces.

The war years. Latvia was under German civil administration from 1941 to 1944, and the *Reichskommissariat Ostland* published statistics, the *Statistische Berichte für das Ostland* of 1942–4 and *Ostland in Zahlen*, that cover crops, livestock, industrial output and employment but little else. Where these fail, the service sectors are interpolated between 1940 and the first post-war observation, and the movement between 1943 and 1945 is taken from the Lithuanian ratios. The estimates for 1941–5 are therefore 'rough' in Feinstein's sense, with a margin of 15–25 per cent, and the 1944 trough is an order of magnitude rather than a measurement (Feinstein

1972, 21). Because the Soviet-period series is extrapolated backwards through the war, an error in the wartime linkage would shift its level in 1937–40, so that the overlap tests the linkage together with the rest of the extrapolation; the spliced series, which follows the interwar series to 1940, is affected only through the 1940 average.

Population and borders

Latvia's borders changed once, when the eastern Abrene district was transferred to the Russian republic in 1944. The series to 1944 refer to the borders of the republic and the series from 1945 to present-day borders. This is the convention of the Maddison Project, which treats income per head as unaffected by a territorial change that moves less than a third of the population (Bolt and van Zanden 2014, 629); the transferred district held a few per cent of the pre-war population at most (Misiunas and Taagepera 1993, 74; Hiden and Salmon 1994, 129–30), and was rural and poorer than the national average, so that the break in income per head is small and no adjustment has been made. The population series rests on the censuses of 1920, 1925, 1930, 1935, 1959, 1970, 1979 and 1989 and the year-end estimates of the Latvian statistical office and the 'Baltic 100' database, with mid-year figures as the average of consecutive year-ends; wartime losses, the loss of some 120,000 ethnic Latvians as refugees and repatriates in 1939–49 (King 1965, 86), the Holocaust and the deportations are taken from the demographic literature, and the Soviet garrison of 1940–1 is added to the resident population. Between 1939 and 1945 the resident population fell by 24 per cent, of which a small part, of the order of two points, reflects the border change (Table A1).

The interwar series and its extension to 1919 and 1940

The interwar series is that of Klimantas et al. (2024), which extrapolates the 1935 benchmark of Norkus et al. (2024) to 1920–39 with sectoral volume indices built from the publications of the Latvian statistical office;[5] the benchmark itself was converted into international dollars by price comparisons with Sweden and Lithuania, of the kind used for Lithuania in 1937 and Estonia in 1938. Two years are added here. For 1919, the first year of the republic's existence, the crop, livestock, railway and employment series were carried back with the data for 1919 where they

5. Bukowski, Kowalski and Wroński object that this extrapolation holds relative prices fixed at their 1935 structure although agricultural–industrial relative prices fluctuated between the wars (Bukowski et al. 2026, 6); the objection is the index-number problem discussed under sensitivity below.

exist and with 1920 ratios where they do not; forestry, fishing, finance and the household services follow the population or the sectors to which they are tied. For 1940, the year of the Soviet occupation, the 1940 yearbook, the Ostland publications and the first Soviet returns provide output and employment for most sectors, imports are assumed equal to their deflated 1939 value, and defence includes the Soviet garrison. The resulting figure for 1919, 1,481 Int$$_{1990}$$, is 37 per cent below the 1913 benchmark of Norkus and Markevičiūtė, and places Latvia, with Lithuania, among the most devastated economies in Europe at the end of the First World War; contemporaries and later historians judged Latvia perhaps the most devastated of all (Norkus 2014, 323; Krūmiņš 2024, 44).

Aggregation, splicing and the overlaps

Each of the two new series is compiled in constant benchmark-year prices and expressed in Int$$_{1990}$$ through its benchmark. The interwar series covers 1919–40 and the Soviet-period series 1937–95, so that the two overlap for four years; the Soviet-period series in turn overlaps for six years with a modern series obtained by backcasting the Maddison Project's 2011 benchmark with the World Bank's constant-price growth rates of GDP per capita, which are used in preference to the Maddison Project's own 1990–2010 growth for reasons given below (Bolt and van Zanden 2025; The World Bank 2024). Table 4 reports the overlaps.

In 1937–40 the two series differ by +1.7, +3.2, −1.9 and −0.7 per cent, all within the 5 per cent margin that Feinstein reserved for 'firm' estimates, and not of one sign. The test is demanding. The two series share no sources, the one resting on the republic's own statistics and the other on Soviet yearbooks and post-war archives; no weights, one using the structure of 1935 and the other that of 1990; and no benchmark, one a price comparison with Sweden and Lithuania, the other a purchasing-power-parity study of the Soviet republics fifty-five years later. That the Soviet-period series, extrapolated backwards over half a century of war, collectivisation and forced industrialisation, arrives within 3.5 per cent of an independently measured level in every year, and within 2 per cent in three of the four, is the strongest evidence available that the net error of the extrapolation is small. It cannot exclude errors that happen to offset, but the two index-number biases discussed below, late base-year weights and fixed value-added weights, would both understate growth and so would both place the backcast 1937–40 level above the

Table 4
Overlapping estimates of Latvian GDP per capita at the two splicing points

Year	Earlier-benchmark series (Int$$_{1990}$$)	Later-benchmark series (Int$$_{1990}$$)	Difference (%)	Feinstein grade
(a) Interwar series (1935 benchmark) and Soviet-period series (1990 benchmark)				
1937	3,031	3,082	+1.7	Firm
1938	3,099	3,198	+3.2	Firm
1939	3,108	3,049	−1.9	Firm
1940†	3,006	2,985	−0.7	Firm
(b) Soviet-period series (1990 benchmark) and adjusted MPD series (2011 benchmark)				
1990	7,965	8,196	−2.9	Firm
1991†	7,385	7,205	+2.4	Firm
1992	6,280	4,959	+21.0	Rough
1993	5,625	4,806	+14.6	Good
1994	5,636	4,994	+11.4	Good
1995	5,710	5,018	+12.1	Good

Notes: Difference = (later − earlier)/earlier in panel (a) and (earlier − later)/earlier in panel (b), so that a positive number always means that the 1990-based series is the higher one. Feinstein's grades: firm, under 5 per cent; good, 5 to under 15; rough, 15 to under 25; conjectural, 25 and over (Feinstein 1972, 21). † Splice year: the final series takes the arithmetic average of the two estimates. *Sources:* see online appendix; adjusted MPD series from Bolt and van Zanden (2025) with 1990–2010 growth replaced by World Bank constant-price growth backcast from the 2011 benchmark (The World Bank 2024).

interwar series; a net understatement of as little as 0.1 percentage point a year over 1940–90 would place it 5 per cent above in every overlap year, which is not observed. Broadberry and Burhop (2007, 323–6) regarded time-series projections as consistent with direct benchmarks when they agree within 10 per cent; the Latvian overlap meets that standard in each of four consecutive years, as the Lithuanian and Estonian series do in 1939–40 and 1937–40 (Klimantas 2026; Klimantas and Klesment 2026). The two series are spliced at 1940 by taking the average of the two estimates for that year, which moves the 1940 figure by less than half a per cent.

The second overlap validates the benchmark. In 1990 the Soviet-period series stands 2.9 per cent below the modern series and in 1991 2.4 per cent above it: Bolotin's purchasing-power parities and the Maddison Project's 2011 ICP round, carried backwards with the national accounts, are two routes that share no data and arrive at the same level within 3 per cent. The series are spliced at 1991, the year of restored independence, by averaging. Thereafter the gap widens to 21.0 per cent in 1992 before narrowing to 12.1 per cent by 1995, because the Soviet-period series continues to value the collapse of Soviet-era output at 1990 weights and misses the new private activities that were emerging unrecorded (Kyn 1997, 260), while the modern

series carries the weights of the surviving branches backwards and records a shallower decline, in line with the view that official data overstated the early-transition fall in output (Havrylyshyn et al. 1999, 7, 11); Hamilton and de Vries's recent reconstruction of the post-Soviet republics likewise extrapolates backwards from the most recent official levels (Hamilton and de Vries 2025, 17). The two probably bracket the true fall; the modern series is preferred from 1992. It is anchored on the Maddison Project's 2011 figure for Latvia and carried back with World Bank growth because the Maddison Project's own path for 1990–2010 implies far slower growth than the national accounts record, so that its 1990 level, 15,806 Int$_{2011}$ or 9,916 Int$_{1990}$, stands 24 per cent above Bolotin's independent estimate; since the two share the 2011 anchor, the whole of that difference lies in the growth attributed to the 1990s, not in the level (Bolt and van Zanden 2025; The World Bank 2024).

The final series and earlier estimates

The 1913 figure is the benchmark of Norkus and Markevičiūtė (2021, 603), taken as published; there are no estimates for 1914–18. Following Bolt and van Zanden (2021, 30), the series is compiled in Int$_{1990}$ and converted to Int$_{2011}$ at the ratio of 1.594 used by Bolt and van Zanden (2025), and both versions are reported, with the population, for 1913 and every year from 1919 to 2024 in Table A1 of the appendix; Table 5 compares it with the earlier estimates at the years for which they exist.

Table 5
Latvian GDP per capita: the new series and earlier estimates, selected years

Year	This paper	Earlier estimates (difference from this paper, %)		
	(Int$_{1990}$)	Rosés and Wolf (2010)	Norkus and Markevičiūtė (2021)	MPD 2023
1913	2,361	—	2,361 (+0.0)	—
1922	1,654	1,929 (+16.6)	1,847 (+11.7)	—
1929	2,351	2,798 (+19.0)	2,347 (−0.2)	—
1938	3,099	4,048 (+30.6)	2,836 (−8.5)	—
1973	8,523	—	—	7,846 (−7.9)
1980	8,867	—	—	8,619 (−2.8)
1985	9,284	—	—	8,954 (−3.6)
1989	8,772	—	—	9,825 (+12.0)
1990	7,965	—	—	9,916 (+24.5)

Notes: MPD figures are published in Int$_{2011}$ and converted at 1.594. The 1913 figure of this paper is that of Norkus and Markevičiūtė, taken as published. *Sources:* Rosés and Wolf (2010, 190); Norkus and Markevičiūtė (2021, 603); Bolt and van Zanden (2025); this paper (online appendix).

For the interwar years the new series lies between the two indirect approximations, above Norkus and Markevičiūtė's and well below Rosés and Wolf's, whose 1938 figure would have made Latvia richer than Finland. For the Soviet period the comparison with the Maddison Project is instructive in a way that differs from the Lithuanian and Estonian cases. The Maddison Project's Latvian levels for 1973–85, carried back from a 1990 level with net material product growth, lie 3–8 per cent *below* the new series, whereas its 1989 level lies 12 per cent above it and its 1990 level 24 per cent above: the Maddison Project records Latvian income per head rising by 5 per cent in 1989 and again in 1990, when every physical indicator compiled here was already falling; the EBRD's index of real GDP likewise rises by 6.8 per cent in 1989 and 2.9 per cent in 1990, whereas Krūmiņš's physical series put 1990 output below that of 1985 (Kyn 1997, 269; Krūmiņš 2017a, 348–50). The discrepancy is not in the benchmark alone but in the growth attributed to the late 1980s, which the net material product series overstated, by about 2 percentage points a year on United Nations estimates from around 1980 (Misiunas and Taagepera 1993, 289).

Limitations and sensitivity

Table 6 lists the sources of bias, the sectors they affect and the maximum impact of each on the level of GDP, with the bounds derived as in the companion studies.

Two deserve comment. The first is the index-number problem of a single late benchmark. A late base assigns to each sector the relative prices of the end of the period, and since the sectors that grew fastest were those whose relative prices fell, late weights understate growth; this is the 'Gerschenkron effect', visible in Bergson's GNP for 1928–37, which grows by 175 per cent at 1928 prices and by 62 per cent at 1937 prices (Feinstein and Thomas 2002, 517). It is far smaller here. The effect is concentrated in the early phases of industrialisation (Jonas and Sardy 1970, 85), and for the post-war Soviet economy it is modest: the CIA's move from 1970 to 1982 prices lowered measured growth over 1951–84 by about half a point a year (Joint Economic Committee 1990, 6–7; Harrison 1993, 156). The bound comes from the data: if 1990 weights understated Latvian growth over 1940–90, the Soviet-period series would arrive in 1937–40 systematically above the 1935-weighted interwar series, and it does not. And the procedure is that of every comparable reconstruction, the CIA's 1982 weights for 1950–90 among them, in

Table 6

Sensitivity of the Latvian series to the main sources of bias

Source of bias	Years	Sectors affected	Bias (%)	Max. impact on GDP (%)
Fixed value-added weights (single deflation)	1919–40	All	±2	±2
Labour productivity and real-wage estimates	1919–40	Industry	±10	±2
Missing annual product-tax data	1919–40	Product taxes	±9	±1
No productivity adjustment in employment series	1919–40	Public services	±10	±2
Population movement as proxy	1919–40	Various service subsectors	±30	±1
Incorrect service-sector evaluation	1940–90	Services (excl. transport)	±10	±4
Missing product-tax data	1940–90	Total	−0.5	−0.5
Fixed value-added weights (single deflation)	1940–90	All	−14	−14
Hidden inflation in trade series	1945–90	Trade, finance and insurance	+59	+2
Unknown Soviet military size	1941–89	Defence	±20	±7
No productivity adjustment in employment series	1940–90	Housing, education, health, science, administration	−147	−10

Notes: 'Bias' is the assumed maximum error in the affected indicator, taken as 10 per cent in either direction unless an independent estimate exists; 'Max. impact' is that error weighted by the affected sectors' share of GDP in the year in which it is largest. A minus sign means that growth in the new series may be understated, so that backcast levels are too high, and vice versa. Fixed value-added weights: Swedish single- versus double-deflated accounts (Schön and Krantz 2012, 19; Lobell et al. 2008, 148–53); hidden inflation: difference between official and estimated real inflation in Soviet retail trade (Howard 1976, 607); product taxes: in 1980–90 the ratio of product taxes less subsidies to value-added was 0.5 per cent in Bulgaria and Ukraine, the only socialist economies with data (The World Bank 2024); service productivity: British service-sector labour productivity rose by 147 per cent between 1948 and 1990 (UK Office for National Statistics 2024). *Sources:* Klimantas (2025, 149–56); this paper.

which Maddison found 'nothing untoward' (Maddison 1998, 312). The second is the fixed ratio of value-added to gross output. Thomas and Feinstein's simulations found that the procedure understated value-added growth by between 0.15 and 1 percentage point a year; the Swedish accounts, which allow single- and double-deflated series to be compared, show an understatement of the deflator of 17 per cent over the three-quarters of a century after 1875 (Schön and Krantz 2012, 19; Lobell et al. 2008, 148–53), and the bound in Table 6, 14 per cent over the half-century, is in the lower part of that range, as befits indices that track output product by product and an economy that, like those of Central and Eastern Europe, could not renew the technological bases of its old branches of industry (Berend 1996, 195–7). The sign of the bias depends on the movement of input relative to output prices, and the Swedish figure is taken as a bound on its magnitude rather than as evidence of its direction in a planned economy. The error

accumulates over the extrapolation and is largest in the earliest years of the Soviet-period series, precisely where the overlap shows the series within 3.5 per cent of an independent level.

Of the remaining biases, the largest in Latvia is defence. The Soviet garrison of 80,000 in 1990 was, in proportion to population, the heaviest in the Baltic republics, and the World Bank still counted some 300,000 military personnel and dependants, 11 per cent of the population, at the time of its 1993 report (World Bank 1993, 157); the garrison's component of public services is interpolated between the 1940 and 1990 totals in step with public-administration employment, with a maximum bias of 7 per cent of GDP in either direction. The trade index uses turnover at comparable prices, the one value series in the reconstruction; Howard's estimates of hidden inflation in Soviet retail trade imply an upward bias of at most 2 per cent of GDP (Howard 1976, 607; Kontorovich 2001, 688–9); in Latvia the only published price index, that of the kolkhoz markets in four cities, rose by 4 per cent a year in 1970–5 and 13 per cent a year in 1975–7, while the official indices recorded none (Misiunas and Taagepera 1993, 221). The employment-based indices for the public services make no allowance for productivity change, which understates service growth if productivity rose, with a maximum effect of 10 per cent of the level. Unregistered activity, put at 10–20 per cent of official GDP in 1989 in the IMF's alternative assumptions for Russia, was of a size that OECD economies also displayed (De Broeck and Koen 2000, 5; Schneider 2000, 90–1). The population of the Abrene district at its transfer in 1944 is not counted in the sources available to us: King speaks only of 'several thousands' of Latvian citizens, and Misiunas and Taagepera, like Hiden and Salmon, of about 2 per cent of the pre-war population (King 1965, 84; Misiunas and Taagepera 1993, 74; Hiden and Salmon 1994, 129–30); the share of about 2 per cent used in the population series is a rounded figure rather than a count, but under the Maddison Project's convention the border change has no bearing on income per head at any plausible value. Taken together, the quantifiable biases imply a margin of about 4 per cent around the interwar levels if they are independent, and of 8 per cent if they all point one way, and of 14 per cent for the largest single bias around the Soviet-period levels, or nearly a quarter in the earliest years if the two same-signed biases were both at their maxima; the overlap shows that the realised error is far smaller. The sub-period growth rates on which the argument of the following sections rests are robust to all of them. The

overlap bounds only the net error over 1940–90, but the arithmetic is simple: even if the largest same-signed bias, a quarter of the level, were concentrated entirely in 1968–89, it would add about one point a year to a rate of 0.7 per cent, against 4.0 per cent in 1950–68. No plausible combination of the biases in Table 6 would raise Latvian growth in 1968–89 to the level of 1950–68, reverse Latvia's position relative to the Soviet average, or alter its ranking among the three Baltic economies in 1938 or 1989.

LATVIA UNDER SOVIET RULE: A BRIEF ACCOUNT

The companion papers have examined the Lithuanian and Estonian series at length; this section reads the Latvian one only far enough to establish what distinguishes it, before the three are compared. Table 7 gives the growth of each sector by sub-period and Table 8 the indices themselves at selected years; the annual series is in the appendix.

Table 7
AVERAGE ANNUAL GROWTH OF SECTORAL VALUE-ADDED VOLUME INDICES, LATVIA (PER CENT)

Sector	1919–29	1929–38	1938–50	1950–68	1968–89	1989–91	1938 level regained in	Peak year
Crops	11.8	5.8	−1.5	1.5	0.3	−10.6	1961	1984
Livestock	3.4	3.8	−5.1	4.0	1.7	−5.5	1966	1988
Forestry and fishing	10.5	0.3	−1.4	4.9	0.2	−9.8	1958	1971
Primary sector	5.7	4.3	−3.3	2.8	1.0	−7.6	1967	1989
Food	—	—	−1.9	7.3	2.1	−4.3	1953	1980
Textiles	—	—	1.7	7.2	0.6	−12.8	1949	1987
Other light industries	—	—	4.4	6.3	−0.8	−3.7	1948	1971
Heavy industries	—	—	5.4	10.3	1.2	−9.1	1948	1986
Industry	20.4	6.8	3.9	8.1	0.8	−8.0	1948	1986
Construction	7.8	2.1	7.4	8.4	−0.5	−9.0	1947	1975
Transport and communications	21.2	−0.5	−1.4	12.2	3.2	−37.6	1952	1988
Trade, accommodation and catering	14.9	3.1	1.4	10.2	4.5	6.5	1949	1990
Finance and insurance	37.3	−1.9	−0.9	7.1	5.5	9.5	1951	1990
Housing	2.2	4.2	−5.7	6.0	2.0	−3.5	1959	1987
Public services	−1.2	0.8	8.0	1.0	0.5	−6.2	1945	1986
GDP	7.4	3.6	1.2	5.2	1.4	−8.5	1948	1986

Notes: compound annual growth between end-point years of constant-price value-added indices (1938 = 100). Industrial branch indices are available from 1938 only. '1938 level regained': first post-war year (1945 or later) in which the index is at or above its 1938 value; 'Peak year': year of the highest index value up to 1990. Public services comprise administration, defence, health and education. Growth in 1938–50 is affected by the wartime trough of 1944, for which Latvian sectoral movements are partly extrapolated from Lithuanian ratios (Section II). The 1989–91 column ends in 1991, the splicing year beyond which the final series follows adjusted Maddison Project data. *Sources:* see online appendix.

Sovietisation began at once. Within a day of the proclamation of Soviet power in July 1940 land, banks and the larger industrial, trade and transport enterprises were declared state property; by the end of the year 1,477 industrial enterprises producing 84 per cent of output had been na-

Table 8

VOLUME INDICES OF SECTORAL VALUE-ADDED IN LATVIA, SELECTED YEARS, 1919–95 (1938 = 100)

Year	GDP	Primary sector	Industry	Construction	Market services	Public services
1919	36	39	9	39	27	105
1929	73	68	55	83	81	93
1938	100	100	100	100	100	100
1945	68	47	27	38	55	200
1950	115	67	159	235	74	253
1955	140	60	293	420	126	219
1960	190	92	433	625	185	207
1968	287	109	645	995	366	305
1975	356	111	814	1,183	499	387
1980	365	109	786	976	634	378
1985	392	127	829	962	677	413
1989	383	135	763	900	745	336
1991	321	115	645	746	592	296
1995	233	63	178	242	745	270

Notes: market services: transport and communications, trade, finance and insurance and housing; public services: administration, defence, health and education. The annual indices are in Tables A2 and A3. *Sources:* see online appendix.

tionalised, the lat was exchanged for the rouble at par although free-market prices in Latvia were a fraction of Soviet ones, and in March 1941 the lat was withdrawn without warning and savings above 1,000 lats annulled (Krūmiņš 2017c, 183–8; Misiunas and Taagepera 1993, 32–3; Terleckas 2022, 290, 294).[6] The military aviation plant that Moscow secretly ordered built in Riga in October 1940, with machine tools requisitioned from VEF and the other large works, foreshadowed the role the republic's industry would play (Krūmiņš 2017c, 194–5). The deportation of some 15,000–15,600 people on 14 June 1941, the German occupation, the Holocaust, which killed some 66,000 Latvian Jews, the mobilisations of 1943–4 and the flight of 1944 reduced the resident population by 24 per cent between 1939 and 1945; King's balance of the decade 1939–49 puts the loss at 570,000 people, a quarter of the pre-war population, only a tenth of it from military operations (Labsvirs 1959, 53–4; King 1965, 81–94; Misiunas and Taagepera 1993, 42, 64). Income per head fell on the present estimates by 28 per cent between 1938 and the trough of 1944, and output by two-fifths, but the figures for 1941–5 are the least secure in the series.

6. Misiunas and Taagepera date the abolition of the Baltic currencies to late 1940 (Misiunas and Taagepera 1993, 33); the lat was declared invalid without warning on 25 March 1941 (Krūmiņš 2017c, 188).

Recovery was fast and, in its sources, revealing. Industry regained its 1938 level in 1948 and GDP in 1948;[7] with a smaller population, income per head regained its 1938 level in 1948 and had exceeded it by a fifth by 1950.[8] The fourth five-year plan built on the machine-building and electrotechnical base of the interwar republic, which had made nearly a third of the Soviet Union's radios in 1939, and on labour imported from the Russian interior: net immigration of 272,000 in 1946–50 accounted for nine-tenths of population growth, and industrial employment doubled between 1945 and 1950 (Eglīte 2015, 46–7; Krūmiņš 2017f, 233–4; Vaskela 2014, 244). Moscow's investment, celebrated in Soviet sources as two billion roubles over the five post-war years, came from Latvia's own revenues: the archival budget accounts show that only about a quarter of the revenue raised in the republic went to its own budget, that military and repressive ministries took nearly half of all spending in Latvia in the late 1940s, and that over 1946–90 as a whole the Union retained 18 per cent of what it collected there, and a further 13.3 billion roubles, 15.7 per cent of all revenue raised there, was spent from the republican budget on Soviet military needs (Krūmiņš 2017f, 251–7; Pazdere 2015, 9; Krūmiņš 2018).[9]

Collectivisation came in 1949 and, as everywhere in the Baltic, by deportation. On 25 March and the days following, more than 40,000 people, most of them farming families, were sent east;[10] the kolkhoz share of agricultural land rose from a tenth at the beginning of 1949 to 83 per cent a year later (Labsvirs 1959, 94), the share of farms collectivised to 85 per cent on the count used in Table 15, and to 93 per cent by the end of December 1949 on Krūmiņš's,[11] and the cooperative dairies and credit societies of the interwar republic, already dismantled, were replaced by procurement prices that fell for meat from 16,640 to 680 roubles a tonne (Krūmiņš 2017f, 245–50; Aizsilnieks 1974, 48). Agricultural output, on the present estimates 59 per cent

7. Misiunas and Taagepera likewise date the recovery of pre-war industrial output in Latvia to 1949 (Misiunas and Taagepera 1993, 113); Mertelsmann, who dates Estonia's recovery to the second half of the 1950s, holds that Latvia followed the same pattern (Mertelsmann 2003, 167–9).

8. Consumption lagged output: urban Latvian families in 1961 still consumed 30–50 per cent less meat, eggs and dairy produce than before the war, and the recovery of pre-war living standards is commonly dated to the 1960s (Misiunas and Taagepera 1993, 185–6; Puur et al. 2019, 38).

9. The budget accounts omit implicit transfers through prices; Aage suggests that with cheap energy counted the Baltic republics 'probably received positive net transfers from Russia', an answer that, as Viksnins had already noted, depends on the price assigned to energy, and that Samonis had rejected for Lithuania (Aage 1997, 102; Viksnins 1986, 252; Samonis 1995, 6–7).

10. Feest's count is 41,445 (Feest 2014, 99–100); for the date see (Strods 2006, 172; Pocius 2019, 75); the older Western literature put the number at 50,000 or more (Labsvirs 1959, 90; Misiunas and Taagepera 1993, 99).

11. Misiunas and Taagepera also give 93 per cent at the end of 1949 (Misiunas and Taagepera 1993, 102); Terleckas gives 90 per cent on 1 January 1950 (Terleckas 2026, 225).

of its 1938 level in 1947, fluctuated between 60 and 67 per cent of it through 1955; grain yields in the early 1950s were little more than half those of the late 1930s, and milk output less than half; the sector regained its pre-war level only in 1967, crops in 1961 and livestock in 1966, and the kolkhoz households' private plots, a tenth of the sown area, still produced more than half of the republic's milk and meat in 1958, while the kolkhoz markets accounted for more than 60 per cent of urban retail sales of meat and milk products (Labsvirs 1959, 219–23, 269–71; Krūmiņš 2017d, 282).

The two decades to 1968 were the fastest the Latvian economy had yet recorded. Industrial output grew by 8.1 per cent a year, heavy industry by 11.9 per cent a year over 1948–69, construction by 8.4 per cent and transport by 12.2; total GDP grew by 5.2 per cent a year and income per head by 4.0. The growth was extensive in the Soviet sense, carried by machine-building for the Union market, by the electric trains, radios and telephone equipment of Riga and by a second wave of chemical and fibre plants in the 1960s, and by labour: a net 292,000 immigrants arrived between 1960 and 1985, and Latvians fell from 62 per cent of the population in 1959 to 52 per cent in 1989 (Krūmiņš 2017e, 303; Eglīte 2015, 50; Puur et al. 2019, 43–4). The one attempt to steer the republic's own development came with the sovnarkhoz reform of 1957, which placed most of Latvian industry under a republican council and raised the share of revenue retained in the republic from a quarter to a half; the 'national communists' who used it to curb heavy industry and immigration were purged in 1959,[12] after which, in the judgement of the literature, the republic's lobbying power in Moscow was weaker than Lithuania's or Estonia's and large all-Union industry dominated (Krūmiņš 2017d, 259–65; King 1965, 203–5; Misiunas and Taagepera 1993, 142–3, 187, 228). Two irregularities in the series fall in these years: a contraction of 3.5 per cent in income per head in 1962, and the turning point of 1968, after which growth never recovered. The first was the Latvian instance of the agricultural crisis that struck the whole Union in 1962–3, when a poor harvest, the restrictions imposed on private plots and livestock since 1958 and the exhaustion of the virgin-lands and maize campaigns forced the leadership to raise the retail prices of meat and butter at the end of May 1962, the only retail

12. Berklāvs was dismissed in July 1959 after the visit of a Moscow commission, Lācis and Kalnbērziņš followed by the end of the year, and the purge ran for two and a half years; Pelše's leadership then lifted the limit on immigration (Smith 2003, 249–50; Strods 2006, 172; Grybkauskas 2022, 487–8).

price rise between 1961 and 1985, to ration bread in many cities and to buy grain abroad in 1963 (Belkharoev 2024, 17–19; Andreenkov 2021, 111–14; Krūmiņš 2017e, 299). On the present estimates the Latvian crop index fell by nearly half in 1962 and did not return to its 1960 level until 1964; Estonia's fell by nearly a fifth, and Estonia's grain deliveries to the all-Union fund fell from 30,000 to 13,000 tons (Purre 1964, 55). Industrial growth paused in the same year.

The years from 1968 to 1989 brought growth slower than any recorded for the communist bloc, on comparator series that, for East-Central Europe, almost certainly overstate it. Income per head grew by 0.7 per cent a year, and after 1973 by 0.2; industry grew by 0.8 per cent a year, and in 1970–85 its growth was the lowest of any Soviet republic (Krūmiņš 2017e, 304). Construction, which had grown by 8.4 per cent a year to 1968, peaked in 1975 and contracted thereafter; on the present estimates the harvests fell into a depression between 1974 and 1982 that has no counterpart in Estonia or Lithuania; and by the mid-1980s the republic consumed nearly twice the electricity it generated and imported feed grain equal to its own production (302, 307). Enterprises of the all-Union ministries produced 34 per cent of industrial output in 1970 and 37 per cent in 1984, and military production, concentrated in VEF, Alfa and Kommutators, employed more than 15 per cent of the labour force in 1985, three times the Estonian share (Guļāns 2015, 34; World Bank 1993, 103–4; Upmalis 2015, 59–61). Latvian income per head, 159 per cent of the Soviet average in 1968 on the present estimates, stood at 136 per cent of it by 1989, and the shops of Latvia were worse supplied from the late 1970s than those of Estonia or Lithuania (Krūmiņš 2017e, 304–5; 2017a, 318; Misiunas and Taagepera 1993, 222; Demuth 2000, 141). The decline began before the Union's: income per head peaked in 1986 and fell by 2.4 per cent a year to 1989, physical output in 1990 was below that of 1985 in most branches (Krūmiņš 2017a, 348–50), and on the present estimates Estonia overtook Latvia durably in 1989, as it had done only briefly before, in the aftermath of each war and in 1962–4.[13]

The transition then removed the Soviet market on which the whole structure rested. Income per head fell by 45 per cent between 1989 and 1993, industry by about two-thirds in three years, and machine-building, which in 1990 employed 132,000 people and exported 92 per cent of its output, only 3 per cent of it to the European Union, saw its two largest branches fall to roughly

13. Official series differ on the last point, still placing Latvian income per head above Estonian in 1988–9 (van Selm 1997, 35; Norkus 2023b, 156).

a quarter of their 1990 output by 1995; the bus and railway-wagon works had closed by 1998 and VEF had been broken up (Šiliņš 2017, 925–9; Soms and Skribāne 2015, 130).[14] Prices were freed in 1991–2, the lats introduced in 1993 and pegged in 1994, and privatisation, slower than in Estonia and Lithuania, had sold about half of former state assets by mid-1998; the banking crisis of 1995, in which Banka Baltija and twelve other banks failed, cost a further 2–4 per cent of GDP (Krastiņš 2017, 811–15; Bitāns and Purviņš 2012, 142; Jeffries 2004, 205–9). Recovery began in 1994, and the 1989 level of income per head was regained in 2004, after Estonia (2001) and Lithuania (2003).

Three things distinguish the Latvian record. The republic entered the Union with the strongest industrial base of the three, specialised most heavily in all-Union machine-building and military production, and surrendered the largest share of its revenue; it lost its only instrument of economic autonomy in 1959; and it stagnated first, declined earliest, and fell furthest in the transition. The comparison that follows asks how much of this was Latvian and how much was Soviet.

THE BALTIC GROWTH PUZZLE: THREE ECONOMIES UNDER THREE REGIMES

With the Latvian series in place, the three Baltic economies can be compared with each other and with the rest of Europe on a common footing for the first time. The comparison is organised around the three regimes under which they have lived since 1918, interwar capitalism, Soviet socialism and restored independence, and around the question that the catch-up literature asks of each: did economies that started poor grow faster than the leaders, and if not, why not? Gerschenkron's late industrialiser borrows technology rather than inventing it and substitutes banks or the state for the prerequisites it lacks, and Solow's model predicts faster growth where capital per worker is scarce; Lewis's surplus labour raises average productivity as it moves off the land; and the cumulative effect of all three widens markets and permits scale economies (Gerschenkron 1962, 8, 14–20, 46–7; Solow 1956, 70–1; Lewis 1954; Abramovitz and David 1996, 22). But Abramovitz and David argued that the potential is realised only where the follower has the 'so-

14. Official series record a deeper fall: the EBRD puts Latvian GDP in 1994 at 52 per cent of its 1989 level (Kukk 2014, 13).

cial capability', the literacy, institutions, managerial competence and commercial organisation, to absorb what it borrows, and where the leader's technology is congruent with the follower's endowments; and Crafts and Toniolo, reading Europe's post-war golden age through that lens, found that even the apparently automatic catch-up of the 1950s and 1960s was conditioned on institutions and policies that fostered investment, technology transfer and the reallocation of resources (Abramovitz and David 1996, 23–35; Crafts and Toniolo 2010, 297–330). The Baltic states are an unusually clean test of these propositions. The human-capital component of their social capability, literacy, was already exceptionally high in the Estonian and Latvian provinces of the Russian Empire, where 85–95 per cent of those aged ten and over could read in 1897 (Raun 2017, 71); its institutional component, the commercial, financial and managerial organisation of the economy, was destroyed and rebuilt twice from outside; and in 1913 all three were among the better-placed peripheral economies of the continent. Tables 9, 10, 11 and 12 summarise the century.

Table 9

Average annual growth of GDP per capita by regime and sub-period, 1913–2023 (per cent)

Period	Latvia	Lithuania	Estonia	Baltics	Finland	Nordics	Western core	Western periphery	ECE	USSR
Interwar capitalism										
1913–38	1.1	0.9	1.1	1.0	2.1	1.9	1.0	1.2	0.8	1.7
1913–29	0.0	−0.2	0.2	0.0	1.6	1.8	1.0	1.4	1.3	−0.1
1929–38	3.1	3.0	2.7	2.9	3.1	2.1	1.1	0.8	0.1	5.0
Soviet socialism (Latvia, Lithuania and Estonia)										
1938–89	2.1	2.7	2.4	2.4	3.1	2.6	2.4	3.0	2.3	2.4
1938–50	1.5	0.9	2.6	1.5	1.4	2.1	0.6	−0.1	0.8	2.4
1950–68	4.0	4.9	3.9	4.3	3.6	3.1	3.6	5.2	3.9	3.4
1968–89	0.7	1.8	0.9	1.2	3.6	2.5	2.4	3.0	1.7	1.5
Restored independence										
1989–2023	2.1	2.7	2.0	2.3	1.2	2.0	1.4	1.5	3.2	1.7
1989–95	−8.9	−6.6	−5.8	−7.2	−0.8	1.6	1.3	2.1	−1.6	−8.0
1995–2007	8.5	7.7	7.7	8.0	3.7	3.8	2.4	2.9	5.7	6.8
2007–23	1.8	2.6	0.9	2.0	0.0	0.7	0.6	0.2	3.2	1.8
1913–2023	1.8	2.3	2.0	2.1	2.3	2.3	1.8	2.1	2.2	2.0

Notes: compound annual rates between end-point years, Int$$_{1990}$. Regional rates are growth rates of population-weighted average GDP per capita. Baltics: Latvia, Lithuania and Estonia; Nordics: Denmark, Finland, Iceland, Norway and Sweden; Western core: Austria, Belgium, Denmark, France, Germany (West Germany 1950–97), the Netherlands, Norway, Sweden, Switzerland and the UK; Western periphery: Finland, Greece, Iceland, Ireland, Italy, Portugal and Spain; ECE (East-Central Europe): Bulgaria, Czechoslovakia, Hungary, Poland, Romania and Yugoslavia; USSR: Soviet Union within 1990 borders, former Soviet Union after 1991. *Sources:* Baltic states: this paper, Klimantas (2026) and Klimantas and Klesment (2026); other series: Bolt and van Zanden (2025), extended to 2023 with The World Bank (2024) as documented in the online appendix.

Table 10

GDP per capita relative to the Western core, the Nordic countries and the USSR, selected years (per cent)

	Percentage of Western core						Percentage of Nordics			USSR
Year	Latvia	Lithuania	Estonia	Baltics	W. periphery	ECE	Latvia	Lithuania	Estonia	Latvia
1913	59	41	51	49	55	39	81	56	71	—
1929	50	34	45	42	58	41	60	41	54	—
1938	60	40	52	49	57	37	66	44	57	144
1950	67	41	66	55	52	38	62	38	61	144
1968	72	51	70	62	69	40	72	51	70	159
1973	67	51	63	59	71	40	67	51	64	155
1989	51	45	52	48	78	35	50	44	51	136
1995	27	28	33	28	82	29	26	27	32	—
2008	52	52	58	53	85	45	43	43	48	—
2019	59	66	65	63	79	57	49	55	54	—
2023	64	69	65	67	81	63	52	56	52	—

Notes: Int$$_{1990}$$; regional aggregates are population-weighted. The last column gives Latvian GDP per capita as a percentage of the Soviet average within 1990 borders; from 1946 the Soviet level is Maddison's series rescaled to Bolotin's 1990 benchmark, to which the Latvian series is anchored (1938 uses Maddison's level and is the less certain). Country groups as in Table 9. *Sources:* as Table 9.

Table 11

GDP per capita in the Baltic states, Finland, the Western core and the USSR, selected years, 1913–2023 (Int$$_{1990}$$)

Year	Latvia	Lithuania	Estonia	Finland	Western core	USSR / former USSR
1913	2,361	1,632	2,058	2,111	4,008	1,414
1929	2,351	1,577	2,113	2,717	4,695	1,386
1938	3,099	2,057	2,685	3,589	5,164	2,150
1950	3,722	2,298	3,667	4,253	5,569	2,586
1968	7,544	5,399	7,348	8,093	10,542	4,735
1973	8,523	6,464	8,080	11,085	12,771	5,515
1989	8,772	7,858	8,958	16,946	17,329	6,473
1995	5,018	5,203	6,277	16,162	18,769	3,933
2008	13,015	13,099	14,604	25,175	25,157	9,266
2019	16,222	18,184	17,883	25,016	27,652	11,060
2023	17,603	19,162	17,837	25,142	27,595	11,482

Notes: USSR within 1990 borders, former Soviet Union after 1991; from 1946 the Soviet level is Maddison's series rescaled to Bolotin's 1990 benchmark, to which the Baltic series are anchored. Western core as in Table 9; the annual Latvian series is in Table A1. *Sources:* as Table 9.

Table 12
GDP per capita as a percentage of the Western core, averages by sub-period, 1919–2023

Period	Latvia	Lithuania	Estonia	Baltics	Western periphery	ECE
1919–28	47	37	47	43	60	43
1929–38	57	39	51	48	59	37
1946–55	67	42	65	55	55	38
1956–67	66	45	65	56	60	38
1968–79	67	51	65	59	71	41
1980–89	59	48	57	54	76	38
1990–94	33	35	38	35	80	28
1995–2007	36	36	45	38	86	34
2008–23	54	59	60	58	80	53

Notes: averages of the annual ratios of GDP per capita, Int$$_{1990}$, to that of the Western core; regional aggregates are population-weighted; 1939–45 are omitted. Country groups as in Table 9. *Sources:* as Table 9.

The puzzle that the long series reveal is this. In 1913 Latvia stood at 59 per cent of the income per head of the Western core and Estonia at 51, levels comparable to Finland's and Italy's; Lithuania, at 41 per cent, exceeded Greece and Portugal. All three were literate, serfdom had ended in Latvia (outside Latgale) and Estonia in 1816–19, two generations before Russia, and within a decade Estonia and Latvia would carry out the most radical land reforms in interwar Europe and Lithuania a more moderate one (Clemens 2003, 30; Norkus 2014, 358–9). Yet by 2023, after a century in which they grew under three different systems, Latvia stood at 64 per cent of the Western core, Estonia at 65 and Lithuania at 69: the region as a whole had moved from 49 to 67 per cent, a gain of eighteen percentage points over 110 years, and against the Nordic countries Latvia had fallen from 81 to 52 per cent. The Western periphery, the group of poorer Western economies to which the Baltic states belonged by income in 1938, rose over the same century from 55 to 81 per cent. Within the region the ranking reversed: Latvia, the richest Baltic economy in 1913, in 1938 and in all but a handful of years to 1988 (Estonia stood marginally above it in the aftermath of each war, 1919–23 and 1947–8, and in 1962–4), is now the poorest, and Lithuania, the poorest for most of the century, is the richest. The sections below take the three regimes in turn.

The First World War and the collapse of the empires gave Lithuania, Latvia and Estonia their first opportunity to build independent states, and left them in ruins. The German advance of 1915 stabilised along the Daugava, within reach of Riga, for two and a half years; the imperial government evacuated Latvia's industry eastwards, and most of it never returned; Riga lost more than half its population between 1914 and 1919 (Kahk and Tarvel 1997, 107; Norkus and Markevičiūtė 2021, 616; Karnups 2017b, 103). Table 13 places the result beside the other belligerents. Between 1913 and 1919 Latvia lost 35 per cent of its population, 59 per cent of its GDP and 37 per cent of its income per head, Lithuania 36, 60 and 38 per cent; of the European economies for which estimates exist, only the Soviet Union lost as large a share of its output. Estonia, untouched by major fighting until 1918, lost less. Pre-war Latvia had been called the 'Belgium of the East' for its industry; by 1919 it resembled Belgium in the devastation it had suffered relative to its size (Hiden and Salmon 1994, 77).

Table 13
Demographic and economic impact of the two world wars (percentage change)

	Population	GDP	GDP per capita
(a) First World War: decline from 1913 to the 1918/19 trough			
Austria	−5	−38	−35
Belgium	0	−32	−32
France	−7	−25	−31
USSR	−1	−60	−59
Lithuania	−36	−60	−38
Latvia	−35	−59	−37
Estonia	−9	−29	−22
(b) Second World War: decline from 1938 to the 1944–6 trough			
Greece	3	−64	−65
Germany	−6	−58	−56
Austria	1	−51	−52
Netherlands	6	−47	−50
Poland	−23	−57	−49
Lithuania	−5	−48	−48
Latvia	−23	−40	−28
Estonia	−20	−42	−29

Notes: each column is measured from the pre-war year to its own trough within the window named. Lithuania in panel (b) within actual borders (the 1939 loss of Klaipėda and gain of Vilnius); within post-war borders its population fell by 22 per cent. *Sources:* Baltic states: this paper and the companion series; others: Bolt and van Zanden (2025).

The 1920s were a decade of reconstruction rather than growth. On the present estimates Lithuania regained its 1913 income per head by 1923, but Estonia only in 1929 and Latvia in 1930: the recovery was protracted in the two northern economies, the most integrated into the Russian market, and slowest in Latvia, which had lost most of its industrial capacity (Aldcroft 2006, 95–6; Crampton 1994, 103).[15] Between 1913 and 1929 income per head in the three countries together was unchanged (Table 9), while the Western core grew by 1.0 per cent a year and the output of Czechoslovakia, one of Berend's 'success stories of the twenties', by 40 per cent (Berend 1998, 243); Latvia's 1922–9 growth of 5.1 per cent a year was bounce-back growth from a very low base, financed by foreign capital, and it stopped when the pre-war level was regained. The new states rebuilt on the land. Agrarian reforms expropriated the estates of the Baltic-German and Polish nobility, which had held between a quarter and a half of all land, and created a class of smallholders; gold-backed currencies were introduced in 1922 (Latvia and Lithuania) and 1928 (Estonia); exports were redirected from Russia to Britain and Germany and shifted from manufactures to butter, bacon, flax and timber, in conscious imitation of Denmark (Kasekamp 2010, 113–15; Norkus et al. 2025a, 1–2; Vaskela 1996, 121; Bernanke 2000, 74; Terleckas 2022, 126, 131). Agriculture still employed 67 per cent of the Latvian labour force in the late 1930s and 63 and 74 per cent of the Estonian and Lithuanian (Table 14).

The 1930s look different. Between 1929 and 1938 income per head grew by 3.1 per cent a year in Latvia, 3.0 in Lithuania and 2.7 in Estonia, against 1.1 per cent in the Western core, 0.8 in the Western periphery and 0.1 in East-Central Europe; among European economies only the Soviet Union, whose figures for the period are contested, and Finland grew as fast, and all three Baltic states closed their gap with the Western core by six to ten percentage points (Table 10).[16] The Depression was shallower and shorter than in most of Europe: Lithuania avoided recession altogether, and Latvia's and Estonia's downturns of 1931–3, of 10 and 8 per cent from peak to trough, were milder than most in Europe though deeper than those of the Nordic countries (3.6–6.5 per cent), which are regarded as among the mildest on the continent, and Latvia's banking

15. On their own series Norkus and his co-authors date the recoveries to 1923 in Lithuania, 1924 in Estonia and 1930 in Latvia (Norkus et al. 2025a, 6).

16. Rosés and Wolf's Maddison-based estimates give Latvia 4.1 per cent a year over 1929–38 against 4.9 for the Soviet Union, and the authors note that the Latvian and Estonian figures may be on the high side (Rosés and Wolf 2010, 188, 190).

crisis of 1931, in which three joint-stock banks failed, was more severe still (Grytten 2008, 370, 372; Bernanke 2000, 91; Terleckas 2022, 236). The Baltic performance has usually been read, by contemporaries and by Aldcroft, as the fruit of export-led agricultural modernisation (Royal Institute of International Affairs, Information Department 1938, 189; Aldcroft and Morewood 2013, 119–20). The sectoral evidence does not support that reading. Trade openness fell sharply after 1929, in Latvia from 38 to 22 per cent of GDP,[17] the processed share of exports fell in both Latvia and Estonia, and the Baltic states remained the most agrarian and least industrial part of Europe outside the Balkans (Table 14). What grew was industry for the home market: Latvian industrial output rose by 6.8 per cent a year over 1929–38, a rate exceeded among capitalist economies only by Lithuania and matched only by Finland, behind the import-substituting tariffs and state enterprises of the authoritarian governments of the 1930s, Lithuania's dating from 1926 and Estonia's and Latvia's from 1934 (Kahk and Tarvel 1997, 113–14, 116; Karnups 2017a, 134, 140), although in Latvia's case the re-industrialisation has been judged more an inadvertent result than a consciously pursued policy (Klimantas et al. 2024, 11, 36). Latvia was the one Baltic economy in which the shares of both agriculture and industry in GDP rose in the 1930s at the expense of services, reflecting heavily subsidised agriculture (Aizsilnieks 1968, 713, 835; Karnups 2017a, 135–6) and of the contraction of financial and public services (Klimantas et al. 2024). The growth of the 1930s was achieved not through international trade but despite its contraction, and Baltic industry, as it had since the 1920s, served the home market (Valge 1994, 188).

Author check (Table 14, agricultural employment c. 1938). Morys's Table 7.7 (Morys 2021, 181) confirms only Czechoslovakia (37.5 per cent in 1930). It gives the Soviet Union 57.8 (1939) against 65 in the table, Bulgaria 71.8 (1930) against 80, Hungary 54.2 (1930) and 50.0 (1941) against 53, and Greece 56.6 (1930) where the table shows '—'. `Datasheets/Sources.csv` names Mitchell and Ivanov (p. 552) for these cells; confirm the source and year of each, and cite Ivanov if retained.

The verdict on the interwar regime is mixed. Over 1913–38 as a whole, Latvian and Estonian income per head grew marginally faster than the Western core's, by 1.1 and 1.1 against 1.0 per cent a year, and Lithuanian marginally slower; the region's position relative to the core, 49 per cent in 1913 and 49 in 1938, did not change. The 1930s were among the most dynamic

17. Period averages of 1925–9 and 1935–8 (Table 14); the single-year ratios of Norkus and his co-authors, built on the same GDP series, fall from 49 per cent in 1929 to 27 in 1938 (Norkus et al. 2025b, 10).

Table 14

Economic structure, industrial growth and openness: the Baltic states and their peers, c. 1937 and c. 1990

	Share of GDP, c. 1937			Share of GDP, c. 1990			Agr. empl.	Ind. growth	Trade openness	
	Agr.	Ind.	Serv.	Agr.	Ind.	Serv.	c. 1938	1929–38	1925/9	1935/8
Lithuania	44	18	38	22	36	42	74	8.0	20	22
Latvia	40	23	37	18	35	47	67	6.8	38	22
Estonia	38	23	40	17	41	42	63	5.7	35	31
Baltics	40	21	38	19	37	44	68	6.9	—	—
Finland	32	30	38	7	34	60	57	6.3	42	34
Iceland	44	16	40	10	25	65	46	3.7	88	49
Ireland	25	24	51	12	46	43	48	4.6	79	50
USSR	31	32	37	19	41	40	65	10.3	—	—
Czechoslovakia	24	35	41	17	48	35	37	−0.8	—	—
Hungary	34	30	37	24	38	38	53	3.6	—	—
Bulgaria	52	15	33	13	57	30	80	3.0	—	—
Portugal	30	27	43	7	38	55	49	2.2	22	20
Greece	54	26	20	14	25	61	—	5.7	22	16
Western core	14	38	49	3	32	65	25	1.1	—	—
Western periphery	36	26	39	8	35	56	50	3.3	—	—
East-Central Europe	37	27	36	14	36	46	65	2.2	—	—

Notes: shares of gross value-added at current prices; 'Ind.' includes construction. c. 1937 is the average of 1935–9 and c. 1990 of 1985–93 where annual data exist, otherwise the nearest year (1990 for the Baltic states and ECE). Regional shares are simple averages. Agricultural employment share and industrial output growth (compound annual rate) from the comparator datasheets of the online appendix, which give the source of each cell; trade openness is exports plus imports as a percentage of GDP, averaged over the years shown; '—' not available. *Sources:* GDP shares: this paper (benchmarks of Table 1), Klimantas and Zirgulis (2020), Klimantas and Klesment (2026), Morys (2021, 179), Palgrave Macmillan Ltd (2013), Amaral (2019, 158), Jonsson (1999, 24), Ivanov (2012, 460), Bolotin (1992, 184), United Nations (1995, 182), Houpt et al. (2010, 346); employment and industrial growth: Palgrave Macmillan Ltd (2013), Morys (2021, 181) (Czechoslovak employment), *Ostland in Zahlen* (1942, 2), Klein et al. (2017, 86); trade: Norkus et al. (2022a, 2022b), Klasing and Milionis (2014), Royal Institute of International Affairs, Information Department (1938, 122, 155), Jonsson (1999, 21).

years the region has known, and by 1938 Latvia and Estonia retained the places among the best-performing peripheral economies of Europe that they had held in 1913, which is what Aldcroft meant in ranking the Baltic states, with Greece, as the most promising performers of Europe's periphery (Aldcroft 2006, 105, 171). But the interwar years also disposed of a belief that has shaped Baltic self-perception ever since: that Latvia and Estonia stood level with Finland. They had been close in 1913, Latvia at 112 per cent of the Finnish level; by 1938 Latvia stood at 86 per cent and Estonia at 75, because Finland grew at 2.1 per cent a year through the two decades while the Baltic economies spent the first of them recovering what the war had destroyed. The

difference was not primarily one of social capability: Latvian literacy, though below Finnish (11 per cent of Latvians aged ten and over were illiterate in 1935, under 1 per cent of Finns aged fifteen and over in 1930), was high for the European periphery, and Latvian university enrolment per head, 4.3 per thousand in the 1930s, was nearly double the Western core's and, in Kalniņš's contemporary comparison, the highest in the Baltic (Klimantas 2025, 25; Norkus 2014, 102–3). It was one of resources and markets. Finland's forests gave it an export staple for which Western demand recovered quickly after 1920 and which carried an engineering sector with it; the Baltic states had lost the only large market their industry had ever served, and the food exports that replaced it faced Danish competition and British quotas.

Soviet socialism, 1940–1991

The Soviet occupation of June 1940, the German occupation of 1941–4 and the renewed fighting of 1944 cost Latvia and Estonia a fifth or more of their population, Lithuania within its actual borders a twentieth, and the three, on the present estimates, between 28 and 48 per cent of their income per head (Table 13). The losses were demographic before they were material: deportations, the Holocaust, war deaths and the flight of some 250,000 people to the West in 1944 removed more than a million people, among them a disproportionate share of the professional and commercial classes; the fall in income per head, 28 per cent in Latvia and 29 in Estonia, was correspondingly smaller than the fall in output, and roughly half the per-capita losses that Table 13 records for Austria, the Netherlands or Poland. Lithuania's larger fall is examined in the companion paper. Recovery was rapid, as it was everywhere in the Soviet sphere: Latvia regained its 1938 level of total GDP in 1948, Estonia its 1939 level in 1948 and Lithuania in 1949, and with smaller populations their income per head regained the 1938 level in 1946, 1948 and 1948 respectively, two years before the Western core or periphery regained theirs. The war had destroyed people rather than plant; with a fifth of the population of the northern republics gone, capacity stood underused, and the early recovery rested on putting labour, increasingly immigrant labour directed from Russia, back to work on existing capital, the mechanism that Vonyó and Markevich identify across post-war East-Central Europe (Vonyó and Markevich 2021, 290; Misiunas and Taagepera 1993, 108).

The command economy was installed within five years. Industry and banks were national-

ised in 1940–1 and again in 1945–7; the republics were incorporated into the all-Union planning system in 1946; and the deportations of 1948–51, the largest single operation being that of March 1949, which took more than 40,000 people from Latvia and some 33,000 and 21,000 from Lithuania and Estonia, broke the resistance of the countryside to collectivisation, which was essentially complete within two years (Table 15) (Krūmiņš 2017c, 183; Feest 2014, 99–100; Misiunas and Taagepera 1993, 102–3, 108). The agricultural indices record the cost. Latvian agricultural output, which had already fallen to 59 per cent of its 1938 level in 1947, hovered between 60 and 67 per cent through 1955, regained the pre-war level only in 1967, twenty-nine years after the pre-war peak, and never exceeded 135 per cent of it before 1991, despite mechanisation and fertiliser; Estonia's recovery came in 1959 and Lithuania's in 1957.[18] Nothing comparable to the Soviet famine of 1932–3 occurred, probably because the Baltic collectivisation did not coincide with a procurement crisis, and the herds survived, Labsvirs noting that it had become 'almost impossible for the kulaks to slaughter their livestock' (Labsvirs 1959, 92); but in every Baltic republic the private plots, on 5 to 10 per cent of the sown land, were still producing a large share of agricultural output, two-fifths in Lithuania in 1975 (Labsvirs 1959, 268–9; Misiunas and Taagepera 1993, 191, 231–2).

What collectivisation did deliver was labour. Industry and construction expanded in the 1950s at rates the capitalist republics had never seen: Latvian industrial output grew by 8.1 per cent a year over 1950–68 and construction by 8.4, Lithuanian industry by 10 per cent a year to 1975. The mechanism was the Soviet one of the 1930s, the forced migration of labour from countryside to town and the extraction of the agricultural surplus for reinvestment (Allen 2003, 99, 176–85, 190, 211). Lithuania, where 80 per cent of the population still lived on the land in 1946, had the labour reserve to sustain it; Latvia and Estonia, more urban and with low fertility, did not, and the planners drew instead on immigration from the Russian interior, which raised Latvia's population by 41 per cent between 1950 and 1989 and reduced ethnic Latvians to barely half of it (Kasekamp 2010, 154–5; Puur et al. 2019, 42–4; Grybkauskas 2011, 205). Each republic was assigned a role in the Union's division of labour: Lithuania food

18. Hiden and Salmon date Estonia's return to the 1938–9 level of agricultural output to the early 1960s, and Norkus and his co-authors put all three republics' return to 1938 levels in the early 1960s (Hiden and Salmon 1994, 131; Norkus et al. 2025a, 15).

Table 15
COLLECTIVISATION OF BALTIC AGRICULTURE, 1947–55

	Latvia		Lithuania		Estonia	
	Farms coll. (%)	Output 1938 = 100	Farms coll. (%)	Output 1938 = 100	Farms coll. (%)	Output 1938 = 100
1947	<1	59	<1	82	0	81
1948	2	67	<1	86	0	90
1949	10	64	4	94	6	86
1950	85	67	62	91	80	85
1951	97	64	89	82	93	83
1952	98	65	>90	76	97	84
1953	>98	61	94	77	99	80
1954	>98	64	>94	83	>99	79
1955	>98	60	>94	78	>99	77
Deported, 1940–53	58,000 (3%)		132,000 (5%)		45,000 (4%)	
of which 1948–53	42,000		100,000		35,000	

Notes: 'Farms coll.': share of farms collectivised at the beginning of the year (mid-year for Estonia in 1952–3); output is the volume index of agricultural value-added (crops and livestock). Deportations in brackets as a share of the population. *Sources:* output: this paper and companion series; collectivisation and deportations: Riekstiņš (2020, 69); Drizulis (1986, 241–2); Museum of the Occupation of Latvia (2006); Girnius (1988, 471); Skebas and Spečiūnas (2018); Taagepera (1979, 265); Õispuu (2014, 12).

processing, light industry and, later, oil refining and nuclear power; Estonia oil shale, electricity and textiles; Latvia machine-building, the electrotechnical and radio industry, railway rolling stock and microbuses, and a large share of all-Union and military enterprises (Misiunas and Taagepera 1993, 184, 234, 238–40; Krūmiņš 2017g, 278, 305; Guļāns 2015, 34). Table 9 records the result. Between 1950 and 1968 income per head grew by 4.0 per cent a year in Latvia, 3.9 in Estonia and 4.9 in Lithuania, faster than the Western core, the Nordic countries, Finland or the Soviet Union as a whole, though not than the Western periphery, whose Iberian and Greek members were in the middle of their own industrialisation. Latvia's position relative to the Western core rose from 67 per cent in 1950 to 72 per cent in 1968, Estonia's to 70 and Lithuania's to 51: the two northern republics stood higher relative to the core than they had in 1913, and the region as a whole stood closer to Western income than at any time before 2019. Against the Union into which they had been absorbed the Baltic republics kept their lead: Latvia entered the Union at 144 per cent of the Soviet average, stood at 159 per cent in 1968 and at 136 per cent in 1989.[19] The gap narrowed only in the last two decades, as Latvia stagnated faster than the Union, and the high income was inherited rather than created by Soviet investment, which, as a

19. Official net material product per head put Latvia 42 per cent above the Soviet average in 1968 and 21 per cent above in 1989 (Misiunas and Taagepera 1993, 185; Norkus 2023b, 156).

share of the republic's GDP, remained below the Soviet average throughout the Cold War on the official data themselves, though per head of population it exceeded the Soviet average for much of the period, including 1946–60 (Klimantas 2025, 192; Krūmiņš 2018; Vaskela 2013, 110).[20]

The two decades of catch-up suggest that the socialist system, whatever its human cost, was a viable instrument of modernisation in the medium term. Vonyó and Markevich call the 1950s and 1960s a golden age of growth for East-Central Europe's planned economies, though one that fell short of the western periphery, and Broadberry and Klein find a similar pattern for Czechoslovakia (Vonyó and Markevich 2021, 281–2; Broadberry and Klein 2011, 44–5). They also mark its limit. The first warning came in 1955, when the industrial expansion of 16–24 per cent a year that all three republics had sustained since 1948 halved and never recovered; the sovnarkhoz reform of 1957–65, which placed industrial management in Riga, Vilnius and Tallinn, was contemporaneous with that deceleration rather than a cure for it, and in Latvia it ended in 1959 with the purge of the 'national communists' who had used the autonomy to slow immigration and favour local consumption (Misiunas and Taagepera 1993, 142–3, 187–9; Krūmiņš 2017d, 262–3; Smith 2003, 249–50). A brief recession in 1962, of 3.5 per cent in Latvia and 1 per cent in Lithuania, with Estonian growth slowing to under 1 per cent, was the Baltic echo of the harvest failure and food crisis that gripped the Union in 1962–3 (Section III), and it passed unnoticed in a historiography attuned to the long upswing. The decisive break came in 1968, less uniformly. Growth rates in all three economies turned down in the late 1960s and, after a respite in the early 1970s, fell until the end of the decade; convergence with the Western core turned negative in Latvia and Estonia in 1968–9, three years after the recentralisation that ended the sovnarkhoz experiment, and at the point at which the Union as a whole ceased to converge on Western levels (Misiunas and Taagepera 1993, 203, 227–8; Drėmaitė 2007, 252). Lithuania, poorer and still endowed with a rural labour reserve, maintained a positive catch-up record until 1975 and briefly again in the early 1980s, carried by food processing, light industry and the construction of its refinery and nuclear plant. Between 1968 and 1989 income per head grew by 0.7 per cent a year in Latvia, 0.9 in Estonia and 1.8 in Lithuania, against 2.4 per cent in

20. According to a 1956 memorandum of the Lithuanian government, investment per head over 1946–55 was 3,800 (pre-1961) roubles in Latvia, 6,500 in Estonia and 2,000 in Lithuania; in 1988 Latvia's investment per head was 95 per cent of the Soviet average, its capital stock per head 114 per cent (Grybkauskas 2011, 148–9; van Selm 1997, 35).

the Western core and 3.0 in the Western periphery. The region that had been Europe's second-fastest-growing in the 1950s and 1960s became its slowest in the 1970s and 1980s, and Latvia, at 0.7 per cent a year over the two decades, grew more slowly than any economy of the communist bloc for which a series exists (there is none for the German Democratic Republic in 1968), on comparator series that, for East-Central Europe, almost certainly overstate growth.

Why did Latvia, the richest of the three in 1968, stagnate first and most completely? Three features of the series stand out. The first is construction. Latvian construction peaked in 1975 and contracted by 1.9 per cent a year to 1989, and Estonian construction stagnated from the same year, whereas Lithuanian construction grew faster than GDP through the 1970s and 1980s; since investment was allocated in Moscow, the pattern suggests that the northern republics, whose labour reserves were exhausted, attracted a shrinking share of the Union's investment, which shifted eastwards to Siberian energy (CIA 1990, 2, 4; Crafts and Toniolo 2010, 323) and, within the Baltic, towards Lithuania (Grybkauskas 2022, 492–3). The second is agriculture: Latvian harvests fell into a prolonged depression between 1974 and 1982 that has no counterpart in Lithuanian or Estonian data and whose institutional rather than climatic character is suggested by that contrast, so that the 2 per cent a year recovery of agricultural value-added in 1979–89 was a major reason why Latvian income per head did not fall in that decade. The third is the structure of industry. Heavy industry, which by 1990 made up 48 per cent of Latvian industrial value-added against 41 per cent in Estonia and 55 in Lithuania, had grown by 11.9 per cent a year in 1948–69 and grew by 1.0 per cent in 1969–89; the machine-building and electrotechnical complexes that served the all-Union market were, by the 1980s, mass-production plants of a generation that Western Europe had already left behind, barred by CoCom controls from the technologies that would have renewed them and by the system's incentives from the innovations its own research institutes produced (Broadberry and Klein 2011, 49; Crafts and Toniolo 2010, 315, 323). Services, the one sector that grew throughout, were growing from a share of GDP that in 1990 was still that of a middle-income economy of the 1930s (Table 14); the shift from agriculture towards industry under planning was not followed by a shift towards services, whose share in East-Central Europe was held up only by government services, while structural change continued in the capitalist periphery (Vonyó and Markevich 2021, 285–6).

By 1987 the system was beyond repair. Latvian income per head declined by 2.4 per cent a year from its Soviet-era peak of 1986, while the Lithuanian and Estonian economies grew by less than 0.2 per cent a year, and in 1989 Latvia, the Baltic leader in all but a handful of years since 1913, ceded its place to Estonia (official net material product per head still placed Latvia marginally ahead, 2,844 roubles against 2,780 (Norkus 2023b, 156)). The Baltic republics had also lost the position they held in 1938 among Europe's peripheries: Greece, the poorest capitalist economy on the Western European mainland, had overtaken all three, and Latvia stood at 51 per cent of the Western core, below its level of 1913, and at 50 per cent of the Nordic countries, where it had stood at 81 per cent in 1913. The progress of the Soviet decades was real, in urbanisation and housing as well as in output (Krišjāne et al. 2019, 226), even if living space per head in Riga had not regained its 1940 level by 1990 (Treija and Bratuškins 2019, 174), and the claim that living standards stagnated under Soviet rule is unfounded; but two decades of growth had been cancelled by two of stagnation, and the measure of the system is not the growth of 1950–68 but its inability to sustain it.

Restored independence, 1991–2024

The transition compressed into four years the deepest contraction in the modern history of the three countries. Latvian income per head fell by 45 per cent between 1989 and 1993, Lithuanian by 36 and Estonian by 34 per cent, falls comparable to those of the Second World War and nearly twice as deep as in East-Central Europe (Table 16). The disintegration of the Union removed at a stroke the specialisation, scale economies, guaranteed demand and cheap inputs on which the Baltic industries had been built, and exposed them to Western competition; entire branches disappeared: Latvia's minibus industry, which had supplied 14,000 to 17,000 vehicles a year to the Union market in the 1980s, had closed by 1998, and in the machine-building sector, which in 1990 accounted for over a quarter of industrial output and nearly 7 per cent of GDP, the two largest branches were by 1995 producing about a quarter of their 1990 output (Klimantas 2025, 39; Krūmiņš 2017g, 305; Šiliņš 2017, 925–9). The literature attributes the depth of Latvia's fall to its heavy concentration of all-Union and military enterprises and to the slowness with which its industrial base was reformed, and the shallowness of Estonia's to an industry smaller in scale, less dependent on Soviet supply and demand, and more quickly reoriented towards

Finland and Sweden; the pattern in Table 16 is consistent with that reading (Norkus 2023a, 164–6; Grennes 1997, 9; Krūmiņš 2024, 115). The depth of the collapse has been disputed: the Maddison Project's series, starting from inflated late-Soviet levels, imply falls of 36–47 per cent, Åslund's adjustments for the shadow economy only 12–21 per cent, and Hamilton and de Vries's backward extrapolation from recent official levels 30–51 per cent. The present estimates, which splice the Soviet-period and modern series at the point of their closest agreement, lie close to Hamilton and de Vries's and well below the Maddison Project's for Lithuania; on the present estimates, as on Hamilton and de Vries's, Latvia's was the deepest of the three Baltic contractions. The official series disagree among themselves: national GNP figures for 1990–3 make Lithuania's fall the deepest (56 per cent, against 48 in Latvia and 39 in Estonia), whereas the EBRD and Economist Intelligence Unit (EIU) series rank Latvia's as the deepest (Aage 1997, 96; Kukk 2014, 13; Kukk 2005, 169).

Table 16
Decline in GDP per capita from 1989 to the 1990s trough: alternative estimates (per cent, trough year in brackets)

	MPD 2023	This paper	Åslund	Hamilton and de Vries
Lithuania	−47 (1994)	−36 (1994)	−21 (1994)	−30 (1993)
Latvia	−46 (1993)	−45 (1993)	−12 (1995)	−51 (1993)
Estonia	−36 (1994)	−34 (1993)	−16 (1994)	−36 (1992)
Baltics	−43 (1994)	−38 (1994)	—	−34 (1993)
ECE	−20 (1992)	—	—	—
USSR	−40 (1996)	—	—	—

Notes: 1989/1991 and 1989/1990 ratios from this paper are applied to Åslund's and Hamilton and de Vries's series, which begin in 1991 and 1990 respectively. 'This paper' for Lithuania and Estonia refers to the companion series. *Sources:* this paper; Bolt and van Zanden (2025); Åslund (2001, 15); Hamilton and de Vries (2025).

Recovery began in 1994–5 (Kukk 2014, 12–13) and, once begun, was faster than anything the region had experienced. Between 1995 and 2007 income per head grew by 8.5 per cent a year in Latvia, 7.7 in Lithuania and 7.7 in Estonia, three times the Western core's rate and well above East-Central Europe's; the 'Baltic tigers' regained their 1989 levels in 2001 (Estonia), 2003 (Lithuania) and 2004 (Latvia), signed association agreements with the European Union in 1995 and joined it in 2004 (Jeffries 2004, 138, 157, 163, 188), and have since adopted the euro and joined the OECD, the only former Soviet republics to do so. The sources of the recovery are those the catch-up literature would predict for a capital-scarce economy with a skilled,

cheap labour force and a legacy of misallocated resources: foreign direct investment, much of it Nordic, and the leap into digital infrastructure (Berend 2009, 123–37; Korhonen 2001, 452, 454). But the recovery was not automatic, as the failure of most other post-Soviet economies to converge attests. It rested on reforms, price liberalisation by 1992, currency boards or fixed pegs by 1994, privatisation and the reorientation of trade to the West, sustained through a fall in income comparable to the Second World War; the two factors usually credited are a national commitment, born of the restoration of independent statehood, that made the costs acceptable, and the anchor of EU accession, whose conditions locked the reforms in, whereas in Belarus, Ukraine or Moldova reform was captured by the old elite (Åslund 2013, 267, 318; Korhonen 2001, 458, 461). The demographic cost was high: between 1989 and 2021 Latvia lost 29 per cent of its population to emigration and natural decrease, Lithuania 24 and Estonia 15 per cent (Klimantas 2025, 42; Hazans 2013, 66; Mežs 2024, 213).

The credit-fuelled boom of 2004–7 ended in the sharpest contraction in the European Union: Latvian income per head fell by 17 per cent between 2007 and 2010, 12.8 per cent of it in 2009 alone, and the internal devaluation with which the country defended its peg has been read both as a model of adjustment and as an excessively painful one (Åslund and Dombrovskis 2011, 1; Blanchard et al. 2013, 326, 350; Krasnopjorovs 2013, 6–7). Between 2011 and the pandemic the three economies grew at 3.3–4.6 per cent a year; Lithuania and Latvia resumed growth afterwards, Estonia did not, and they no longer move together. Lithuania, having overtaken Latvia durably in 2010 (on the present estimates it had stood above it in the transition years 1992–9) and Estonia in 2018,[21] is now the richest of the three and still converging; Estonia's GDP contracted in each year from 2022 to 2024 (Povilauskas 2025, 12), its income per head has fallen relative to the European Union average (Tsaurkubule and Zhukovskaya 2024, 159), and on the present series it stood in 2024 below its 2019 level; Latvia, which grew by 1.8 per cent a year over 2007–23 against Lithuania's 2.6 and Estonia's 0.9, remains the poorest of the three and in 2023 stood at 64 per cent of the Western core, where Lithuania stood at 69 and Estonia at 65 per cent. Measured against the European Union as a whole, Lithuania is now markedly closer to the

21. The pre-2010 ranking is source-dependent: Eurostat's purchasing-power-standard series place Lithuania above Latvia in 2000, 2004, 2010 and 2017, while the Maddison Project keeps Latvia ahead in 1993–9, where the World Bank's PPP series do not (Starkevičiūtė 2013, 98; Puur et al. 2019, 39; Norkus 2014, 73–4).

average than it was in 1938, Estonia is roughly where it was, and Latvia, at about three-quarters of the EU average against about 90 per cent in 1938, when the average included the poor interwar East, has still not regained the relative position it held before the war; measured against the Nordic countries, no Baltic state has regained its position of 1913, and part of the catch-up that has occurred is the arithmetic of a shrinking population: Latvia's total GDP relative to the Western core was no higher in 2023 than in 1990 (Klimantas 2025, 205). Latvia shows the symptoms of a middle-income trap, rising labour costs without a corresponding rise in productivity and investment, that the OECD's surveys have documented (OECD 2022, 70; 2024, 17, 23), while the Latvian debate has traced Latvia's lag to state capture by 'redistribution industries', a less cohesive elite and the loss of human capital (Krastiņš 2023, 48–50), explanations that remain untested.

Convergence tests

Table 17 tests the three regimes against the simplest version of the catch-up hypothesis, the unconditional convergence regression of growth on initial income across the countries of Europe, for the periods 1913–38, 1938–89 and 1989–2019, whose end-points are the peak years of prewar capitalism, interwar capitalism, state socialism and modern capitalism before the pandemic; the observations for each economy are in Table A4.

In the interwar period the slope is gently negative, −0.32 percentage points of annual growth for each unit of log initial income, and most of the poor economies with negative residuals are the East-Central European ones, including the Baltic states, that would later adopt state socialism; Latvia and Estonia lie close to the line and grew faster than the Western core, while Lithuania fell well below it. The Cold War period shows two regimes rather than one. Among capitalist economies convergence was pronounced, and every member of the Western periphery outgrew the core; among socialist economies the fitted line lies below the capitalist one at every level of income. Four of the ten, the poorest, Lithuania, Bulgaria, Yugoslavia and Romania, outgrew the Western core's 2.4 per cent a year; Estonia and the Soviet Union matched it; and the remaining four, Latvia among them, fell short. Measured against the socialist line alone, however, Latvia lies on it and Estonia above it, so that both performed at least as well as their group. The post-communist period restores a single downward-sloping line, of −0.56, and almost every

Table 17

Growth and initial income in Europe under three regimes: unconditional convergence regressions

	(a)	(b) Cold War, 1938–89			(c)
	1913–38	All	Socialist	Capitalist	1989–2019
Slope	−0.32	−0.54	−0.87	−0.99	−0.56
	(0.20)	(0.17)	(0.26)	(0.13)	(0.34)
Intercept	3.61	7.04	9.03	10.92	7.13
R^2	0.09	0.27	0.59	0.78	0.06
Residual standard deviation	0.6	0.5	0.3	0.3	1.1
Number of economies	30	29	10	19	43
Western-core growth	1.02		2.40		1.57
Growth (residual from the column's regression line)					
Latvia	1.09 (−0.03)	2.06 (−0.61)	2.06 (+0.01)	—	2.07 (+0.05)
Lithuania	0.93 (−0.31)	2.66 (−0.23)	2.66 (+0.25)	—	2.84 (+0.75)
Estonia	1.07 (−0.09)	2.39 (−0.36)	2.39 (+0.21)	—	2.33 (+0.32)

Notes: ordinary least squares regressions of the compound annual growth of GDP per capita (per cent) on the natural logarithm of initial GDP per capita, Int$$_{1990}$, across European economies; standard errors in parentheses. Socialist economies in column (b): Albania, Bulgaria, Czechoslovakia, Hungary, Poland, Romania, the USSR, Yugoslavia and the three Baltic states; Germany is omitted from (b). Spain enters (a) and (b) with 1935 in place of 1938 because of the civil war. One observation per economy: before 1989 the Soviet Union, Yugoslavia and Czechoslovakia enter and their member or successor states do not, except the three Baltic states; from 1989 the successor states enter and the aggregates do not. Residuals are in percentage points a year. The data for each economy are in Table A4. *Sources:* this paper, the companion series and Bolt and van Zanden (2025).

peripheral economy outgrew the core; Lithuania lies well above the line, Estonia above it and Latvia on it, behind Poland, Romania, Albania and Hungary, the frontrunners of the current convergence. The regressions are descriptive, not causal. The samples are small and the slopes imprecise (Table 17 gives standard errors); measurement error in initial income, of the size the margins of Section II imply, biases the slope towards zero and moves individual residuals; and the Cold War regression, which starts in 1938 for economies that became socialist only in 1945–8, confounds the regime with war losses that ranged from a tenth to two-thirds of income per head (Table 13), so that the framework cannot establish a causal effect of communist rule (Klimantas 2025, 210). Read with those limits, the conclusion is threefold. Under neither interwar capitalism nor Soviet socialism did any Baltic economy exceed the convergence expectation set by the European pool; under socialism all three fell below it, Latvia furthest, as did every socialist economy except Yugoslavia; and under restored capitalism Lithuania has exceeded the expectation, Estonia has exceeded it by less, and Latvia has merely met it.

Sources of catch-up and decline

The lack of consistent input data precludes growth accounting, but the series permit a reading of the three episodes of catch-up, the 1930s, the 1950s and 1960s and the years since 1995, and of the two reversals that separated them. The interwar catch-up was the closest the region came to Gerschenkronian automaticity: the Baltic economies were among the least industrialised in Europe, so that the industrial growth of the 1930s came from a low base and from the imitation of simple technologies, Danish cattle and Western electronics; capital was so scarce that modest investment yielded large returns; and structural backwardness insulated them from a Depression that struck the industrial economies hardest. But the episode was also conditioned. Governments mobilised savings for targeted investment; the land reforms had produced unusually egalitarian distributions of land (Kõll 1994, 48) and societies in which status was earned rather than inherited, the 'social, political and cultural victory' of the modernisers that Rostow placed at the start of sustained growth (Rostow 1959, 7); and literacy and Western legal traditions, strongest in Latvia and Estonia, eased the absorption of borrowed technology. The autarkic environment of the 1930s set the limit: whether these advantages could have been converted into sustained convergence in the open post-war order is the counterfactual that history did not run.

The socialist catch-up of 1950–68 was driven by the four automatic mechanisms at once, technology transferred from an already industrialised Union, capital deepening from a low base, labour moved off the land, and scale economies within a vast internal market, and it was enabled by two conditions: the social capability inherited from the interwar years, which made Latvia and Estonia, with their skilled workforces and intact infrastructure, the natural recipients of Soviet mass-production technology; and the coercion of a system that could collectivise agriculture, direct labour and investment across borders without friction, and import workers by the hundred thousand. The reversal after 1968 reflects the exhaustion of those conditions and the character of the system. By the 1970s, on one reading, mechanisation and industrialisation in the Baltic republics had caught up with Soviet levels and the advantages in social capability that had distinguished Latvia and Estonia within the Union had eroded (Klimantas 2025, 217); the labour reserve of the northern republics was spent (Misiunas and Taagepera 1993, 236–7), and investment was redirected towards Lithuania (Grybkauskas 2022, 492–3). What remained was

the Soviet institutional system itself, which, as Crafts and Toniolo argue for the Union as a whole, wasted capital, failed to reward innovation and could not move from mass production to flexible production and the service economy (Crafts and Toniolo 2010, 314–15, 323), the changes that were transforming the capitalist periphery across the Gulf of Finland; and, as Vonyó and Klein have shown for East-Central Europe, it also failed to sustain the capital formation its own statistics claimed (Vonyó and Klein 2019). Having converged on the rest of the Union in social capability and probably in capital stock, and having no control over policy, the Baltic republics most likely shared in the collapse of the Soviet growth model for the reasons that apply to the Union as a whole; that Latvia, which had converged furthest and had the heaviest concentration of all-Union industry, stagnated first is consistent with that reading, but the republic-specific policies of Moscow remain to be analysed before it can be shown to follow from it.

The modern catch-up has again drawn on the advantages of backwardness, in the form of the structural distortions and obsolete capital the Soviet era left behind, cheap and educated labour and the opening of capital and trade flows; but it has been more plainly conditional than either of its predecessors, resting on the reforms of the 1990s and the discipline of EU accession, and it has been uneven in a way that the automatic mechanisms cannot explain. Latvia, which in 1938 had the strongest prerequisites of the three for Western convergence, has gained least from the regime that rewards them most, and is now, on the present series by a narrow margin, the poorest. The Baltic century thus underscores the proposition with which the catch-up literature has ended: the benefits of backwardness are realised only through institutions that enhance social capability, and they can be forfeited, as Latvia's trajectory from the richest Baltic economy in 1913 to the poorest in 2024 shows, by regimes that destroy its institutional component and by policies that fail to rebuild it.

CONCLUSION

This paper has constructed the first annual, internationally comparable series of Latvian GDP per capita for the Soviet period, 1937–95, from physical output, transport and employment data anchored to Bolotin's 1990 benchmark, and joined it to the published interwar series, extended to 1919 and 1940, and to adjusted modern data, to give a continuous series for 1913 and 1919–2024 in 1990 and 2011 international dollars. Built from different sources, with different weights

and from benchmarks fifty-five years apart, the interwar and Soviet-period series agree within 3.5 per cent in every year of their 1937–40 overlap, and within 2 per cent in three of the four, and the Soviet-period series meets a modern series derived from the 2011 ICP round within 3 per cent at the 1990–1 splice. The series supersedes the Maddison Project's figures, which carry a benchmark inflated by the first year of transition backwards with net material product growth and so place Latvian income in 1990 a quarter too high while understating it in the 1970s, and it is ready for the Maddison Project Database. With the Lithuanian and Estonian series it completes the national accounts of the Baltic states and provides, for the first time, modern GDP estimates for three republics of the Soviet Union; the procedure can be applied to the others.

The Latvian series tells a story of inherited strength spent. Latvia lost more than a third of its population and three-fifths of its output in the First World War and spent the 1920s regaining its 1913 income; it grew by 3.1 per cent a year in the 1930s behind import-substituting industrialisation, lost a quarter of its population and 28 per cent of its income per head in the Second World War, recovered by 1948, and grew by 4.0 per cent a year between 1950 and 1968 as a machine-building and electrotechnical workshop of the Soviet Union, built with immigrant labour on the skills and industrial traditions of the pre-1914 decades and the interwar republic (Vaskela 2013, 115–16; Misiunas and Taagepera 1993, 184; Puur et al. 2019, 42–3). From 1968 it stagnated earlier and more completely than its neighbours, at 0.7 per cent a year to 1989, in years in which investment was redirected, construction contracted, harvests fell and the all-Union industries aged; it ceded the Baltic lead to Estonia in 1989, suffered the deepest transitional collapse of the three Baltic states, regained its 1989 income only in 2004, and was overtaken by Lithuania in 2010.

The comparison of the three Baltic economies across three regimes yields four conclusions. First, under interwar capitalism the region recovered rather than converged: the dynamic 1930s only restored the relative position of 1913, and the parity with Finland that Baltic memory assumes had already been lost by 1938 (Vilpišauskas 2014, 228; Norkus 2023b, 154). Second, Soviet socialism delivered two decades of genuine catch-up, faster than the Western core though slower than its periphery, through coercive mobilisation of labour and capital on a base of inherited social capability; and then two decades in which the region grew more slowly than any

other in Europe, as the mechanisms of extensive growth were exhausted and the system proved unable to replace them. The gains of 1950–68 were cancelled by 1989, and the Baltic republics, among the best-placed peripheral economies of Europe in 1938, were poorer than Greece in 1989. Third, restored independence has brought the only sustained convergence of the century, at a demographic cost without parallel in post-socialist Europe, and it has been conditional on reform and uneven in outcome: Lithuania has exceeded the convergence expectation of the European pool, Estonia has exceeded it by less, and Latvia, which in 1938 was best placed of the three, has merely met it. Fourth, in the convergence regressions of each period no Baltic economy exceeded the expectation set by the European pool before 1989, and all three fell below it under socialism; against the Nordic countries none has regained its relative position of 1913.

The puzzle with which the paper began has, then, a partial answer. The Baltic states did not fail to converge for want of social capability; they possessed its human-capital component throughout, and lost half a century in which it could not be used, first in the autarkic and unstable environment of the interwar period, which limited what egalitarian and literate agrarian societies could achieve, and then in a system that could exploit their capability for a generation but not renew it. What they have made of their third opportunity differs, and the divergence since 2008 between Lithuania and its northern neighbours, which the Latvian series brings into focus, is the question the comparison leaves open. The series presented here pose that question, and the Soviet-era turning points of 1955 and 1968, with a precision that was not available before. Answering them will require what the series cannot provide: republican investment data corrected for Soviet pricing, of the kind Vonyó and Klein have built for East-Central Europe (Vonyó and Klein 2019); a Soviet-period benchmark with republican rather than Union-wide prices; and consumption-side estimates against which the output-side record of Soviet living standards can be tested.

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APPENDIX

This appendix reports the annual series on which the paper rests. Table A1 gives Latvian GDP per capita in 1990 and 2011 international dollars and the mid-year population for 1913 and every year from 1919 to 2024; Tables A2 and A3 give the constant-price volume indices of sectoral value-added for 1919–95 from which the GDP series of 1919–91 is built, with 1938 equal to 100; Table A4 gives the initial income and growth of each European economy in the convergence regressions of Table 17. The indices for 1992–5 are those of the Soviet-period series and are superseded, in the final series, by the modern series from 1991. The online appendix, available as an Excel workbook, contains the same series together with the 1935 and 1990 benchmarks by sector, the overlapping estimates at the two splicing points, the product-by-product composition of the industrial index, the comparator series used in the tables with their sources, and a source list for every series by year, keyed to the sections A.1–A.7 referred to in Table 2. The series to 1944 refer to the borders of the republic and the series from 1945 to present-day borders.

Table A1
Latvian GDP per capita in 1990 and 2011 international dollars and mid-year population, 1913 and 1919–2024

Year	Int$$_{1990}$$	Int$$_{2011}$$	Pop. (000)	Year	Int$$_{1990}$$	Int$$_{2011}$$	Pop. (000)	Year	Int$$_{1990}$$	Int$$_{2011}$$	Pop. (000)
1913	2,361	3,763	2,275.0	1954	4,345	6,926	1,952.9	1990	7,965	12,696	2,663.2
1919	1,481	2,361	1,480.0	1955	4,304	6,861	1,981.0	1991	7,295	11,629	2,650.6
1920	1,522	2,426	1,629.5	1956	4,472	7,129	2,018.2	1992	4,959	7,904	2,614.3
1921	1,671	2,663	1,715.7	1957	4,640	7,397	2,053.7	1993	4,806	7,660	2,563.3
1922	1,654	2,637	1,786.1	1958	4,812	7,671	2,073.2	1994	4,994	7,960	2,520.7
1923	1,899	3,027	1,814.4	1959	5,238	8,349	2,092.0	1995	5,018	7,998	2,485.1
1924	1,943	3,097	1,835.1	1960	5,476	8,729	2,121.0	1996	5,206	8,298	2,457.2
1925	2,085	3,323	1,850.5	1961	5,668	9,034	2,152.7	1997	5,722	9,122	2,432.9
1926	2,095	3,340	1,862.7	1962	5,466	8,713	2,181.6	1998	6,143	9,791	2,410.0
1927	2,151	3,428	1,875.3	1963	5,699	9,084	2,210.9	1999	6,364	10,144	2,390.5
1928	2,152	3,431	1,887.0	1964	6,149	9,801	2,240.6	2000	6,790	10,823	2,367.5
1929	2,351	3,747	1,896.3	1965	6,470	10,313	2,265.9	2001	7,313	11,657	2,337.2
1930	2,537	4,044	1,905.4	1966	6,688	10,660	2,283.2	2002	7,923	12,629	2,310.2
1931	2,314	3,689	1,916.2	1967	7,210	11,493	2,301.2	2003	8,674	13,826	2,288.0
1932	2,286	3,644	1,927.6	1968	7,544	12,026	2,323.6	2004	9,495	15,136	2,263.1
1933	2,496	3,979	1,938.0	1969	7,666	12,220	2,343.2	2005	10,627	16,940	2,238.8
1934	2,722	4,340	1,946.5	1970	8,045	12,824	2,359.2	2006	12,009	19,143	2,218.4
1935	2,776	4,425	1,953.5	1971	8,244	13,141	2,376.4	2007	13,312	21,219	2,200.3
1936	2,797	4,458	1,960.4	1972	8,328	13,275	2,395.7	2008	13,015	20,746	2,177.3
1937	3,031	4,831	1,967.7	1973	8,523	13,585	2,415.8	2009	11,345	18,084	2,141.7
1938	3,099	4,940	1,982.8	1974	8,617	13,735	2,437.2	2010	11,067	17,641	2,097.6
1939	3,108	4,955	1,987.6	1975	8,834	14,081	2,456.1	2011	11,561	18,428	2,059.7
1940	2,996	4,776	1,984.3	1976	8,952	14,270	2,471.0	2012	12,528	19,970	2,034.3
1941	2,891	4,608	1,892.0	1977	8,900	14,187	2,485.1	2013	12,912	20,582	2,012.6
1942	2,703	4,308	1,799.6	1978	8,780	13,995	2,497.9	2014	13,280	21,169	1,993.8
1943	2,596	4,138	1,806.6	1979	8,737	13,927	2,506.0	2015	13,914	22,179	1,977.5
1944	2,236	3,565	1,646.9	1980	8,867	14,134	2,511.7	2016	14,375	22,913	1,959.5
1945	2,744	4,374	1,518.6	1981	9,002	14,350	2,519.4	2017	14,990	23,894	1,942.2
1946	2,886	4,600	1,635.2	1982	8,945	14,258	2,531.1	2018	15,709	25,040	1,927.2
1947	3,028	4,826	1,786.6	1983	9,103	14,509	2,546.0	2019	16,222	25,859	1,913.8

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Table A1 (continued)

Year	Int$$_{1990}$$	Int$$_{2011}$$	Pop. (000)	Year	Int$$_{1990}$$	Int$$_{2011}$$	Pop. (000)	Year	Int$$_{1990}$$	Int$$_{2011}$$	Pop. (000)
1948	3,290	5,245	1,871.6	1984	9,346	14,897	2,562.0	2020	15,968	25,453	1,900.4
1949	3,490	5,564	1,885.4	1985	9,284	14,799	2,578.9	2021	16,775	26,739	1,884.5
1950	3,722	5,932	1,887.0	1986	9,438	15,044	2,599.9	2022	17,077	27,220	1,879.4
1951	3,874	6,175	1,894.3	1987	9,124	14,544	2,626.6	2023	17,603	28,059	1,881.8
1952	4,009	6,391	1,905.7	1988	9,049	14,424	2,653.8	2024	17,725	28,253	1,871.9
1953	4,089	6,518	1,926.0	1989	8,772	13,982	2,667.4				

Notes: 1913 from Norkus and Markevičiūtė (2021) as published; 1919–39 from the interwar series of Klimantas et al. (2024) extended to 1919 (Section II); 1940 and 1991 are averages of the two overlapping series; 1941–90 from the Soviet-period series of this paper; 1992–2024 from Bolt and van Zanden (2025) with 1992–2010 growth from The World Bank (2024), extended to 2024 with World Bank and Eurostat data. Int$$_{2011}$$ = Int$$_{1990}$$ × 1.594. Population within the borders of the republic to 1944 and present-day borders thereafter; mid-year averages of year-end figures. *Sources:* see online appendix.

Table A2

Volume indices of sectoral value-added in Latvia, 1919–95: primary and secondary sectors (1938 = 100)

Year	Crops	Live-stock	Forestry, fishing	Primary	Food	Textiles	Other light	Heavy	Industry	Constr.
1919	19.8	51.0	35.8	39.2	—	—	—	—	8.6	38.9
1920	22.9	54.4	39.0	42.5	—	—	—	—	13.7	42.0
1921	44.1	59.4	43.3	53.2	—	—	—	—	21.3	48.8
1922	41.3	57.7	59.6	51.9	—	—	—	—	26.0	55.1
1923	49.2	65.5	73.6	60.1	—	—	—	—	35.5	58.6
1924	50.3	64.2	70.6	59.6	—	—	—	—	38.2	61.9
1925	63.1	66.2	76.7	65.5	—	—	—	—	42.1	64.5
1926	51.6	69.3	97.7	64.3	—	—	—	—	42.9	65.8
1927	51.1	70.4	114.0	65.5	—	—	—	—	46.6	64.5
1928	38.6	70.4	119.6	61.3	—	—	—	—	48.9	74.1
1929	60.2	71.2	96.9	68.4	—	—	—	—	55.1	82.7
1930	73.3	74.4	69.1	73.8	—	—	—	—	61.9	112.6
1931	54.4	81.4	87.9	72.1	—	—	—	—	52.9	89.9
1932	73.6	82.7	108.9	80.6	—	—	—	—	48.2	72.1
1933	85.2	82.0	121.1	84.9	—	—	—	—	63.6	68.6
1934	102.8	87.6	105.7	93.8	—	—	—	—	73.6	79.4
1935	96.8	94.7	104.9	95.9	—	—	—	—	78.6	72.1
1936	80.2	95.6	117.8	91.1	—	—	—	—	82.4	87.6
1937	102.9	94.0	105.9	97.7	—	—	—	—	93.7	103.1
1938	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
1939	106.3	100.5	98.0	102.5	102.0	88.3	95.7	102.8	100.8	112.8
1940	103.6	82.5	73.9	89.6	98.9	85.3	86.9	79.7	99.4	109.1
1941	97.9	67.5	120.6	80.7	86.9	73.4	65.0	59.7	80.6	110.8
1942	87.2	55.4	90.0	69.1	74.5	39.9	47.3	30.4	54.3	71.3
1943	74.0	53.7	102.2	62.6	69.9	33.6	46.1	28.4	50.3	73.3
1944	66.2	44.6	102.5	54.5	47.5	24.1	35.8	22.0	12.3	19.9
1945	58.4	37.6	61.1	46.6	27.7	14.5	32.4	19.7	27.0	37.6
1946	63.5	35.7	74.0	48.3	40.0	34.7	53.5	30.8	45.7	65.0
1947	82.2	39.0	86.2	58.6	45.1	49.7	89.7	60.0	70.5	100.1
1948	92.8	45.3	86.0	66.6	60.6	80.7	116.7	110.5	105.3	146.4
1949	81.1	50.3	91.5	63.9	69.5	100.8	151.3	157.5	136.6	201.3
1950	83.8	53.6	84.7	66.6	79.6	122.6	167.1	188.9	158.8	234.8
1951	75.9	55.9	91.5	64.3	90.0	161.3	200.4	222.3	192.1	269.0
1952	78.8	55.9	81.5	65.3	97.7	175.1	225.0	263.0	216.5	299.0
1953	70.7	54.9	77.8	61.1	103.6	185.8	260.4	303.3	242.8	346.4
1954	71.0	59.9	85.0	63.9	112.9	201.5	331.3	348.3	284.0	399.2
1955	62.4	59.9	86.8	60.0	120.4	215.4	318.5	375.1	292.9	419.5
1956	76.9	63.4	86.8	68.3	129.7	228.5	318.7	417.9	310.3	454.8
1957	78.6	67.9	97.8	71.6	140.8	244.0	352.6	465.7	340.8	484.2
1958	73.9	77.3	102.5	74.2	151.4	257.8	352.6	524.1	362.7	521.3
1959	93.2	84.5	106.7	86.6	164.7	275.3	386.6	591.3	399.4	581.1
1960	99.8	88.3	113.9	91.7	183.3	286.0	411.9	658.0	432.7	625.1
1961	102.0	89.7	113.7	93.4	183.9	291.9	429.6	700.1	451.0	655.3
1962	53.6	87.4	122.5	70.5	189.1	301.0	431.3	760.0	471.0	664.3
1963	71.3	83.7	139.5	77.1	206.2	313.2	438.0	789.7	488.8	684.6
1964	100.1	79.4	160.4	88.5	233.7	334.1	461.5	877.4	532.2	762.5

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Table A2 (continued)

Year	Crops	Live-stock	Forestry, fishing	Primary	Food	Textiles	Other light	Heavy	Industry	Constr.
1965	95.0	95.7	163.2	94.3	255.1	349.2	457.9	943.9	558.2	822.8
1966	92.1	105.3	169.5	97.9	259.0	368.4	460.3	990.8	577.7	863.4
1967	105.0	106.7	175.4	104.6	272.8	401.8	481.6	1,054.6	614.3	945.0
1968	110.5	108.5	202.1	108.6	283.6	429.3	499.2	1,110.1	645.1	995.3
1969	94.5	110.5	211.7	102.5	305.2	445.9	506.2	1,178.6	675.5	1,019.1
1970	117.9	111.3	241.8	114.3	322.3	486.3	520.3	1,222.5	707.6	1,045.3
1971	106.6	117.0	261.7	112.4	337.5	441.2	530.2	1,328.9	728.9	1,113.5
1972	105.2	122.1	238.0	113.8	358.9	459.8	506.8	1,354.4	739.1	1,052.3
1973	119.1	116.9	228.4	117.3	370.9	474.0	497.5	1,430.3	762.9	1,101.3
1974	100.5	126.3	212.3	113.1	366.3	472.6	491.3	1,506.1	778.8	1,123.6
1975	92.5	129.0	210.5	110.8	393.3	486.6	484.6	1,606.4	813.7	1,183.1
1976	95.6	132.3	209.2	113.8	418.1	483.1	476.8	1,619.1	819.9	1,146.1
1977	91.3	135.8	205.9	113.5	417.9	480.7	468.9	1,611.2	814.6	1,107.4
1978	73.9	132.9	184.3	103.7	426.1	477.1	455.6	1,552.2	796.4	1,068.8
1979	91.7	125.1	187.0	108.0	415.6	473.9	426.6	1,527.4	777.0	993.0
1980	82.1	134.5	199.3	108.6	458.7	471.5	406.5	1,547.6	786.5	975.8
1981	91.9	133.4	193.4	112.3	448.6	470.4	435.0	1,553.5	794.1	986.5
1982	97.3	131.0	195.0	113.6	456.3	465.6	438.0	1,568.7	799.6	952.6
1983	105.3	141.5	206.4	122.7	447.8	469.9	452.6	1,565.4	802.4	957.9
1984	122.1	147.0	211.1	133.2	445.4	480.7	467.7	1,587.1	815.3	994.8
1985	101.4	153.2	208.7	126.8	447.3	488.6	479.0	1,615.2	829.0	961.8
1986	113.7	155.8	220.2	133.9	435.1	498.1	484.8	1,660.8	842.1	992.1
1987	99.3	158.0	223.9	128.5	420.0	512.4	471.6	1,595.7	821.4	965.2
1988	104.0	159.0	222.3	131.2	428.8	501.2	460.8	1,571.5	810.8	941.3
1989	116.6	155.2	212.8	134.8	443.3	483.5	421.5	1,438.3	762.6	900.2
1990	100.0	146.3	185.2	122.2	440.8	422.9	371.5	1,303.1	693.6	793.5
1991	93.2	138.7	173.0	115.1	405.7	367.5	390.8	1,188.6	644.8	745.7
1992	95.3	116.8	112.8	103.7	254.8	237.7	251.1	578.5	365.7	450.0
1993	92.8	90.9	122.8	90.0	151.8	93.4	168.8	350.8	211.4	296.3
1994	78.0	69.1	135.6	72.8	140.9	94.2	155.8	315.2	195.5	269.7
1995	61.4	63.8	157.9	63.2	139.3	87.3	172.0	235.8	177.8	242.0

Notes: industrial branch indices from 1938 only. 1941–4 movements partly interpolated; 1944 from Lithuanian 1943–5 ratios (Section II). Indices for 1992–5 are those of the Soviet-period series and are superseded in the final series by the modern series from 1992 (1991 being the average of the two). *Sources:* see online appendix.

Table A3
Volume indices of sectoral value-added in Latvia, 1919–95: services and GDP (1938 = 100)

Year	Transp., comm.	Trade	Finance	Housing	Public adm.	Defence	Health	Educ., culture	Other	GDP
1919	15.3	19.0	5.0	55.5	59.2	237.2	32.6	31.4	65.2	35.7
1920	28.7	23.4	5.4	59.8	63.8	237.2	35.2	45.2	70.3	40.4
1921	44.0	31.3	5.7	63.2	67.4	158.1	50.5	61.0	76.3	46.6
1922	59.1	38.2	10.3	65.4	69.7	87.8	56.8	73.2	81.2	48.1
1923	71.0	51.5	15.2	67.3	68.7	87.9	61.5	74.5	84.9	56.1
1924	74.2	55.2	37.6	68.8	67.8	93.2	68.6	76.1	87.6	58.0
1925	71.0	60.8	90.0	69.5	71.2	104.1	66.2	78.8	89.4	62.8
1926	76.9	61.0	109.5	70.0	74.6	109.0	68.8	80.8	90.9	63.5
1927	88.8	62.4	118.5	69.8	74.8	106.6	71.7	82.6	91.5	65.6
1928	102.0	65.1	108.2	67.1	76.9	110.6	76.8	83.6	92.6	66.1
1929	104.9	75.9	118.3	69.1	80.0	114.6	82.0	85.9	94.3	72.6
1930	108.1	81.4	158.7	71.6	82.3	118.7	85.6	90.3	95.7	78.7
1931	82.7	68.5	184.8	68.6	83.8	112.7	88.7	89.7	95.9	72.2
1932	70.9	59.3	140.2	82.6	80.5	106.8	91.1	89.8	96.5	71.7
1933	82.8	70.2	128.9	82.8	83.1	100.9	94.6	91.9	96.9	78.7
1934	90.8	79.1	128.9	84.5	87.0	101.0	95.9	90.3	97.2	86.2
1935	88.9	82.4	108.9	85.1	88.9	101.2	95.1	93.0	97.8	88.3
1936	90.1	82.8	97.4	100.1	93.5	100.5	96.8	96.9	98.9	89.2
1937	112.8	96.9	90.2	88.7	98.2	100.2	97.8	98.7	99.4	97.1
1938	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
1939	107.1	99.5	104.2	85.6	100.4	99.1	100.7	96.4	99.0	100.6
1940	101.3	95.0	115.9	74.0	99.3	192.1	99.5	108.8	97.2	97.1
1941	70.2	83.7	82.2	69.1	110.3	218.7	99.9	86.3	123.6	89.6
1942	64.6	63.0	77.6	64.2	121.4	245.3	91.7	83.5	149.5	79.7
1943	52.0	51.6	73.0	59.2	132.5	271.9	92.0	80.7	175.5	76.9

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Table A3 (continued)

Year	Transp., comm.	Trade	Finance	Housing	Public adm.	Defence	Health	Educ., culture	Other	GDP
1944	42.1	41.6	8.4	54.3	86.2	243.4	92.4	57.5	107.4	60.4
1945	38.0	71.2	63.8	49.4	154.6	325.1	85.1	108.8	227.4	68.3
1946	58.0	80.7	81.7	67.0	159.2	334.7	107.4	130.6	297.9	77.3
1947	63.3	90.2	86.8	59.9	163.7	344.2	129.6	152.3	368.5	88.6
1948	70.8	99.7	91.9	56.4	168.2	353.8	151.9	174.1	439.1	100.9
1949	81.7	109.2	97.0	52.9	172.8	363.4	174.2	195.9	524.6	107.9
1950	84.2	118.7	89.4	49.4	177.3	372.9	196.5	217.6	610.2	115.1
1951	94.1	138.8	115.5	52.9	168.6	354.4	221.5	226.6	648.7	120.3
1952	104.9	151.9	122.3	56.4	159.8	336.0	236.1	235.6	681.3	125.2
1953	115.3	181.6	141.7	63.5	151.0	317.5	247.8	244.5	666.8	129.1
1954	125.2	223.1	168.8	70.5	142.2	299.0	259.4	253.5	693.6	139.1
1955	136.3	239.7	158.5	77.6	133.4	280.5	284.1	262.4	777.7	139.7
1956	148.3	256.3	152.6	83.2	123.5	259.8	305.4	275.2	870.4	147.9
1957	163.7	303.8	152.5	88.8	113.7	239.1	314.8	288.1	1,070.7	156.2
1958	175.5	322.8	155.1	94.5	113.7	239.1	331.8	294.5	1,010.8	163.5
1959	185.9	348.9	157.6	100.1	113.7	239.1	357.3	320.1	1,118.5	179.6
1960	209.0	389.2	169.4	105.8	106.1	223.1	382.8	320.1	1,393.6	190.3
1961	251.6	401.1	168.7	112.8	113.7	239.1	399.8	345.7	1,623.9	200.0
1962	303.4	416.5	169.5	116.3	113.7	239.1	399.8	384.1	1,776.4	195.4
1963	350.2	433.1	190.3	116.3	121.3	255.0	399.8	396.9	2,114.3	206.5
1964	392.8	450.9	214.8	119.9	128.8	270.9	408.3	416.1	2,392.4	225.8
1965	437.0	503.1	252.6	123.4	136.4	286.9	433.9	428.9	2,607.6	240.3
1966	468.2	553.0	268.6	123.4	144.0	302.8	433.9	448.1	2,607.6	250.2
1967	590.4	611.1	287.6	134.0	159.2	334.7	442.4	460.9	2,793.0	271.9
1968	663.8	683.4	309.5	140.3	166.7	350.6	450.9	476.9	2,975.1	287.3
1969	690.2	738.5	321.4	145.2	174.3	366.6	459.4	492.9	3,157.2	294.4
1970	716.1	805.0	337.7	151.6	181.9	382.5	467.9	505.7	3,334.7	311.1
1971	742.1	862.0	437.4	155.1	189.5	398.4	476.4	518.5	3,669.7	321.1
1972	781.2	916.2	445.7	158.6	197.0	414.4	489.2	531.3	4,004.7	327.0
1973	808.5	951.4	447.7	169.2	197.0	414.4	500.2	537.7	4,247.1	337.4
1974	800.9	996.0	545.5	172.7	212.2	446.2	510.4	563.3	4,582.1	344.2
1975	841.8	1,051.2	571.4	176.3	219.8	462.2	527.4	576.1	4,824.4	355.6
1976	958.9	1,107.2	604.7	179.8	212.2	446.2	544.5	588.9	4,927.7	362.5
1977	990.6	1,141.5	626.2	183.3	204.6	430.3	561.5	601.7	5,077.4	362.5
1978	1,113.0	1,183.3	652.0	186.8	197.0	414.4	578.5	614.5	5,227.1	359.4
1979	1,138.6	1,227.0	679.0	193.9	189.5	398.4	595.5	627.3	5,331.8	358.8
1980	1,168.9	1,294.5	719.4	197.4	189.5	398.4	608.3	692.0	5,372.3	365.0
1981	1,202.5	1,333.3	739.2	197.4	194.0	408.0	614.2	697.7	5,420.1	371.7
1982	1,178.3	1,289.3	719.0	197.4	198.6	417.6	623.6	702.8	5,408.2	371.0
1983	1,212.2	1,320.4	747.1	197.4	203.1	427.1	631.2	711.8	5,583.2	379.8
1984	1,258.9	1,372.1	784.5	197.4	207.7	436.7	638.9	722.0	5,593.7	392.4
1985	1,250.6	1,423.9	811.5	197.4	212.2	446.2	647.4	731.6	5,602.7	392.4
1986	1,291.8	1,438.2	809.8	200.9	212.2	446.2	677.2	748.9	5,746.2	402.1
1987	1,315.9	1,452.4	819.7	218.6	174.3	366.6	692.5	777.7	5,610.1	392.8
1988	1,322.9	1,580.6	859.7	215.0	159.2	334.7	702.7	822.5	5,626.7	393.6
1989	1,294.3	1,708.7	952.2	211.5	128.8	270.9	680.6	826.4	5,393.4	383.5
1990	791.7	1,851.1	1,070.5	204.5	128.8	270.9	652.5	810.4	5,188.3	347.6
1991	503.7	1,938.2	1,142.4	196.9	147.2	209.3	633.3	754.2	4,908.4	320.8
1992	524.3	2,036.2	1,632.0	164.1	227.0	153.9	671.7	810.4	4,584.5	269.1
1993	668.4	2,036.2	1,632.0	151.5	269.9	94.8	767.6	754.2	4,195.3	236.3
1994	832.8	2,156.0	1,958.4	154.0	306.8	54.9	710.1	738.2	4,162.1	232.8
1995	825.0	2,079.8	3,264.0	143.9	343.6	34.4	710.1	834.5	4,455.9	232.5

Notes: as Table A2. Defence in 1940 includes the Soviet garrison admitted in 1939–40. *Sources:* see online appendix.

Table A4

Initial GDP per capita and growth of the European economies in the convergence regressions

	(a) 1913–38		(b) 1938–89		(c) 1989–2019	
Economy	Initial	Growth	Initial	Growth	Initial	Growth
Albania	—	—	—	—	2,477	3.69
Armenia	—	—	—	—	6,603	0.55
Austria	3,465	0.11	3,559	3.04	16,360	1.72
Azerbaijan	—	—	—	—	5,283	2.38
Belarus	—	—	—	—	7,461	1.60
Belgium	4,220	0.54	4,832	2.47	16,744	1.45
Bosnia and Herzegovina	—	—	—	—	4,210	1.68
Bulgaria[s]	1,137	1.11	1,499	2.83	6,216	2.17
Croatia	—	—	—	—	8,757	1.76
Cyprus	823	1.65	1,238	4.02	9,230	2.44
Czech Republic	—	—	—	—	8,800	2.81
Czechoslovakia[s]	2,096	1.17	2,803	2.26	—	—
Denmark	3,912	1.56	5,762	2.29	18,261	1.70
Estonia[s]	2,058	1.07	2,685	2.39	8,958	2.33
Finland	2,111	2.15	3,589	3.09	16,946	1.31
France	3,485	1.00	4,466	2.69	17,300	1.17
Georgia	—	—	—	—	8,884	−0.38
Germany	3,648	1.26	—	—	16,558	1.92
Greece	1,455	2.47	2,677	2.64	10,111	1.32
Hungary[s]	2,098	0.95	2,655	1.89	6,903	3.07
Iceland	2,015	1.34	2,813	3.70	17,975	1.37
Ireland	2,736	0.44	3,052	2.52	10,880	4.21
Italy	2,545	0.82	3,125	3.25	15,969	1.11
Latvia[s]	2,361	1.09	3,099	2.06	8,772	2.07
Lithuania[s]	1,632	0.93	2,057	2.66	7,858	2.84
Luxembourg	6,485	0.54	7,425	2.23	22,862	1.37
Macedonia	—	—	—	—	6,403	1.00
Malta	704	2.02	1,160	3.84	7,909	3.28
Moldova	—	—	—	—	6,454	−1.24
Montenegro	—	—	—	—	4,566	3.50
Netherlands	4,049	1.04	5,250	2.29	16,695	2.00
Norway	2,770	1.74	4,262	2.88	18,157	3.66
Poland[s]	1,739	0.91	2,182	1.90	5,684	3.97
Portugal	1,250	1.35	1,747	3.55	10,372	1.78
Romania[s]	1,101	−0.36	1,006	2.71	3,940	4.46
Russia	—	—	—	—	8,009	2.30
Serbia	—	—	—	—	6,878	1.12
Slovakia	—	—	—	—	7,989	2.59
Slovenia	—	—	—	—	12,445	1.41
Spain	1,922	0.99	2,385	2.96	11,518	2.17
Sweden	2,874	2.16	4,901	2.53	17,524	1.63
Switzerland	5,127	0.65	6,028	2.47	20,901	2.12
USSR[s]	1,414	1.69	2,150	2.37	—	—
Ukraine	—	—	—	—	6,325	0.02
United Kingdom	5,152	0.79	6,266	1.91	16,414	1.35
Yugoslavia[s]	973	1.00	1,249	3.19	—	—

Notes: initial: GDP per capita in the first year of the period, Int$$_{1990}$$ (the regressions use its natural logarithm); growth: compound annual growth of GDP per capita, per cent. [s] socialist economy in column (b) of Table 17. Spain: 1913–35 and 1935–89. '—': not in the sample for the period. *Sources:* as Table 17.